

Evaluation of Project Performance in Participatory Budgeting

Niclas Boehmer¹, Piotr Faliszewski², Łukasz Janeczko²,
Dominik Peters³, Grzegorz Pierczyński⁴, Šimon Schierreich⁵,
Piotr Skowron⁴, and Stanisław Szufa²

¹Harvard University

²AGH University, Kraków

³CNRS, LAMSADE, Université Paris Dauphine - PSL

⁴University of Warsaw

⁵Czech Technical University in Prague

Abstract

We study ways of evaluating the performance of losing projects in participatory budgeting (PB) elections by seeking actions that would have led to their victory. We focus on lowering the projects' costs, obtaining additional approvals for them, and asking supporters to refrain from approving other projects: The larger a change is needed, the less successful is the given project. We seek efficient algorithms for computing our measures and we analyze and compare them experimentally. We focus on the GREEDYAV, PHRAGMÉN, and EQUAL-SHARES PB rules.

1. Introduction	2
2. Preliminaries	3
3. The Measures and How to Compute Them	5
3.1. Cost-Reduction Measure	5
3.2. Add-Approvals Measures	6
3.3. Add-Singletons Measure	7
3.4. Rivalry-Reduction Measure	8
4. The Measures in Practice	8
4.1. Behavior and Correlation	9
4.2. Wieliczka's Green Million	12
5. Future Work	13
6. Bibliography	14
A. Additional Material for Section 2	17
B. Additional Material for Section 3	17
C. Additional Material for Section 4	27

1. Introduction

The idea of participatory budgeting (PB) is to let members of a local community—such as a city or its district—decide how a certain budget B should be spent [Cabannes, 2004, Goel et al., 2019, Rey and Maly, 2023]. To this end, some members of the community first submit projects, together with the costs of implementing them, and then the whole community votes on which of them should be funded, typically by indicating which projects they approve (but see, e.g., the work of Fairstein et al. [2023] for an analysis of other ballot formats). The funded projects are selected using the GREEDYAV rule, which looks at the projects one by one, starting with those that received the most approvals, and selects a project if its cost is within the still-available budget. For example, consider a PB instance with projects a, b, c, d, e , 10 voters, and budget $B = 10$ (we consider a fairly small city), where the project costs and the votes are as follows:

proj.	cost	voters									
		x_1	x_2	x_3	y_1	y_2	y_3	z_1	z_2	z_3	z_4
a	7	✓	✓	✓	✓	✓	✓	-	-	-	-
b	4	-	-	-	-	-	-	✓	✓	✓	✓
c	3	-	-	-	✓	✓	✓	-	-	-	-
d	2	-	✓	✓	-	-	-	-	-	-	-
e	2	✓	-	-	-	-	-	-	-	-	-

In particular, project a costs 7 units and is approved by 6 voters (x_1, x_2, x_3 and y_1, y_2, y_3). GREEDYAV first considers project a (as it is approved by the largest number of voters) and selects it. Then it looks at project b (approved by 4 voters, costing 4 units), but it does not select it as there are only 3 units of budget left. Next it considers project c and selects it, using up the whole remaining budget.

When the city announces the results, it may wish to present additional information to explain why projects lost or won. In particular, proposers and supporters of losing projects may wish to know how close their project was to winning. A city could do this by publishing an “information package” for each losing project. In the example, for the losing project b , the package could note that the project would have won had it been cheaper by one unit, or had it been supported by two or three additional voters (depending on how ties are broken). Having such an information package would make the whole process more transparent.

The GREEDYAV rule has the disadvantage that it frequently underserves minorities (indeed, in our example voters $z_1, \dots, z_4$ do not approve any winning projects, even though they form 40% of the electorate). Researchers have proposed new PB rules, such as EQUAL-SHARES [Peters and Skowron, 2020, Peters et al., 2021] and PHRAGMÉN [Brill et al., 2017, Los et al., 2022], that aim at proportional representation of the voters and avoid this problem. These rules are more involved than GREEDYAV, making it important to explain the outcome to participants. We study performance measures like the ones mentioned above for PBs using these rules, to help produce good information packages. This will be particularly useful for EQUAL-SHARES, which was recently used in real-life PB elections in Wieliczka (Poland) and Aarau (Switzerland).

While the measure based on reducing projects’ costs applies directly to proportional rules, there are some challenges in adapting the measure of how many more approvals are necessary for funding. The reason is that under GREEDYAV it only matters *how many* approvals we add, but under proportional rules it is also important *where* we add them. We illustrate this issue using EQUAL-SHARES, which roughly works as follows: Given n voters and budget B , the rule assigns B/n units of budget to each voter. Then, it proceeds in rounds, where in each round a group of voters *buys* a project if (1) they jointly have sufficient funds, and (2) they are the largest group that can afford a project at this point of time; the voters from this group share the cost of the

project equally among themselves.¹ In our example, each voter obtains one unit of budget, and in the first round voters z_1, z_2, z_3, z_4 buy project b (they are the largest group that can afford a project; project a has more approvals, but its supporters do not have sufficient funds). Then, voters y_1, y_2 , and y_3 buy project c and, finally, voters x_2 and x_3 buy project d . The final outcome is $\{b, c, d\}$, under which only voter x_1 is left without an approved winning project. Now let us see how this outcome would change if e got some additional approvals. If it got approvals from x_2 and x_3 , then EQUAL-SHARES would certainly select it instead of d . Yet, if e got two additional approvals from z_1 and z_2 then it would still lose; these voters would spend all their funds on b and would not help e .

We address this issue with the adding-approvals measure by considering (a) the smallest number of approval additions needed for a project’s victory (for the right selection of voters who add the approvals), (b) the smallest number of approval additions that suffice for a victory no matter which voters perform them, as well as (c) a variant based on randomization and (d) a variant based on adding new voters.

Finally, under a proportional rule, a project may also lose because it is supported by voters for whom the rule chooses other projects that they approve. Indeed, this is exactly why EQUAL-SHARES would not select project e even if it were additionally approved by z_1 and z_2 . If these voters did not approve b , their support for e would suffice for its victory. Knowing that a project could have won if some of its supporters refrained from voting for other projects is useful both for voters to understand how the rule works, and for project proposers for planning their election campaigns.

Technical Contribution. We are interested in two aspects of our measures. First, as real-life PB instances can include hundreds of projects and tens of thousands of voters, we seek fast algorithms for computing them in practice. As some of our measures are NP-hard to compute, we find FPT algorithms for them that work well in realistic settings.

Second, we compute our measures for projects from real-life PB instances from Pabulib [Faliszewski et al., 2023]. We find that measures of the same type are strongly correlated, but that measures of different types may disagree about how well a project performed. All measures provide helpful insights, which we demonstrate by constructing an example information package based on the Pabulib vote data from the 2023 Wieliczka “Green Million” PB, where EQUAL-SHARES was used for the first time.

Proofs of all statements and much additional information on our experiments can be found in the appendix.

2. Preliminaries

A *participatory budgeting* (PB) instance $E = (P, V, B)$ consists of a set of projects $P = \{p_1, \dots, p_m\}$, a set of voters $V = \{v_1, \dots, v_n\}$, and budget $B \in \mathbb{N}$. Each voter v has a set $A(v) \subseteq P$ of projects that he or she approves (referred to as his or her *approval set*), and each project $p \in P$ has a price for implementing it, denoted $\text{cost}(p)$. We extend this notation so that for a project p , $A(p)$ is the set of voters who approve it, and we refer to $|A(p)|$ as the approval score of project p . A set $S \subseteq P$ of projects is feasible if its cost, denoted as $\text{cost}(S) = \sum_{p \in S} \text{cost}(p)$, is at most B . A PB rule is a function that given a PB instance outputs a feasible set S of selected projects (i.e., our rules are resolute and, so, give unique outcomes). We refer to projects in S as the *selected* (or, *funded*) ones, and to the remaining ones as *losing*.

¹If there are voters who do not have enough funds left to pay their full share, they pay the fraction that they have left, but they also count as a fraction of a voter toward the group’s size.

We consider three PB rules, GREEDYAV, PHRAGMÉN [Brill et al., 2017, Los et al., 2022], and EQUAL-SHARES [Peters and Skowron, 2020, Peters et al., 2021]. Each of them starts with an empty set of projects W , performs a sequence of rounds, where in each round it extends W with a single project, and eventually outputs W as the final outcome. Whenever they encounter an internal tie (i.e., two or more projects that fulfill a given condition), they break it using a given, prespecified order over the projects. Our rules work as follows, letting $E = (P, V, B)$ be the input PB instance:

greedyAV (AV). In each round, GREEDYAV considers project p with the highest approval score, that it has not considered yet. If $\text{cost}(W) + \text{cost}(p) \leq B$ (i.e., if it can afford to fund p) then it includes p in W . The rule terminates upon considering all the projects.

Phragmén (Ph). The voters start with empty virtual bank accounts, but receive funds in a continuous manner, one unit of funds per one unit of time. As soon as there is a project $p \in P \setminus W$ such that the voters in $A(p)$ have $\text{cost}(p)$ funds in total and $\text{cost}(W) + \text{cost}(p) \leq B$, the rule includes p in W and resets the bank accounts of the voters from $A(p)$ to zero (these voters *buy* the project). The process stops when for every project p with at least one approval it holds that $\text{cost}(W) + \text{cost}(p) > B$.

Equal-Shares (Eq). This rule also uses voters' virtual bank accounts, but it initiates them to $B/|V|$ per voter and does not provide further funds. Each round proceeds as follows, where b_i is the current account balance of voter $v_i \in V$. The idea is to select a project that its supporters can afford, and such that each of them has to cover as small a fraction of its cost as possible. Formally, a project $p \in P \setminus W$ is affordable if there is $q(p) \in [0, 1]$ such that:

$$\sum_{v_i \in A(p)} \min(b_i, q(p) \cdot \text{cost}(p)) = \text{cost}(p).$$

For each voter $v_i \in A(p)$, we let $q_i(p)$ be the fraction of $\text{cost}(p)$ that v_i needs to cover; it is $q(p)$ if $b_i \geq q(p) \cdot \text{cost}(p)$ (i.e., if v_i can afford its full share) and it is $b_i/\text{cost}(p)$ otherwise. The rule selects an affordable project p with the smallest value of $q(p)$, includes it in W , and charges each $v_i \in A(p)$ with $q_i(p) \cdot \text{cost}(p)$ (if there are no affordable projects, then the rule terminates).²

A well-known issue of EQUAL-SHARES is that it is not exhaustive, i.e., upon termination there still may be sufficient budget left to fund more projects. Hence, in practice one needs to apply one of several *completion methods*. In most of our experiments we do the following: When EQUAL-SHARES terminates, we start the PHRAGMÉN rule, but with voters' bank accounts initiated to their value at the end of EQUAL-SHARES. We call this rule EQ/PHRAGMÉN. In Wieliczka and Aarau, a different completion method was used, which, unfortunately, is too computationally intensive for our full set of experiments (see Appendix A).

Each of the above-described PB rules can be computed in polynomial time using a *round-based* algorithm, which executes each round following the definition. Many of our measures can be computed while running a round-based algorithm, by performing some additional steps in each round.

Definition 2.1. Let f be a PB rule. Let $\text{measure}_E(p)$ be a function that takes as input a PB instance E and project p , and let $t: \mathbb{N} \rightarrow \mathbb{N}$ be some function. We say that $\text{measure}_E(p)$ can be

²In the language of Brill et al. [2022], this definition is based on *cost utilities*. The definition from the introduction matches the formal definition if we define the weighted size of the voter group that approves p as $\sum_{v_i \in A(p)} q_i(p)/q(p)$. Each voter that contributed the full share counts as 1 in this sum, but those who contributed less count as respective fractions. One can verify that EQUAL-SHARES selects a project approved by a group with the largest weighted size.

measure	AV	PH	EQ
cost-red	along/ $O(1)$	along/ $O(1)$	along/ $O(1)$
optimist-add	along/ $O(1)$	along/ $O(n \log n)$	along/ $O(n \log n)$
50%-add	along/ $O(1)$	sampling	sampling
pessimist-add	along/ $O(1)$	NP-com./FPT	FPT
singleton-add	along/ $O(1)$	along/ $O(1)$	brute-force
rival-red	sampling	sampling	sampling

Table 1: Summary of our algorithms for computing the performance measures. By along/ $O(1)$ and along/ $O(n \log n)$ we mean that the measure can be computed alongside the rule, with $O(1)$ or $O(n \log n)$ additional cost per round (where n is the number of voters). By sampling, we mean algorithms based on simulating a given action a number of times. By FPT, we mean the algorithm from [Theorem 3.6](#). By brute-force, we mean adding singleton voters one by one.

computed alongside $f(E)$ at a cost of $t(|E|)$ per round if it is possible to compute its value using a round-based algorithm for $f(E)$, extended to perform at most $t(|E|)$ additional computational steps in each round.

3. The Measures and How to Compute Them

In this section, we describe our measures and provide ways of computing them. While we also analyze their worst-case computational complexity, our focus is on obtaining practically usable algorithms that can be applied to PB elections from Pabulib [\[Faliszewski et al., 2023\]](#). We focus on GREEDYAV (AV), PHRAGMÉN (PH), and EQUAL-SHARES (EQ) as this allows for a clean presentation. By combining algorithms for EQ and PH, one can obtain algorithms that work for EQ/PHRAGMÉN. We summarize our algorithms in [Table 1](#).

The common feature of our measures is that they correspond to specific actions that either the voters or the project proposers could have taken in the election. In this respect, they are closely related to the margin-of-victory [\[Magrino et al., 2011, Cary, 2011, Xia, 2012\]](#) and, more broadly, bribery notions [\[Faliszewski et al., 2009, Faliszewski and Rothe, 2015, Yang, 2020\]](#). In particular, we borrow ideas from the work of [Faliszewski et al. \[2017\]](#) about bribery in multiwinner elections.

Except for rivalry reduction (see [Section 3.4](#)), our measures are well-defined in all but a few pathological cases (such as a project without any approvals). We list these issues in [Appendix B.1](#).

3.1. Cost-Reduction Measure

Our conceptually simplest measure is the one based on reducing the project’s cost. The measure was also studied by [Baumeister et al. \[2021\]](#). Formally, we define it as follows.

Definition 3.1. *Let f be a PB rule, let $E = (P, V, B)$ be a PB instance, and let $p \in P \setminus f(E)$ be a losing project. We define the cost-reduction measure of p in E , denoted $\text{cost-red}_E^f(p)$, to be the largest value such that if we replace p ’s cost with it, then f selects p .*

That is, $\text{cost-red}_E(p)$ is the project’s cost after the smallest possible reduction that gets p funded (we drop the superscript denoting the rule when it is clear from the context).

For AV, PH, and EQ, it is immediate that we can compute the cost-reduction measure in polynomial time using binary search, but it would require recomputing the rules multiple times. Instead, we compute it alongside our rules, for each round finding the largest cost at which our

project can be selected right then, and outputting the largest of these values. A similar algorithm is used by [Baumeister et al. \[2021\]](#) for AV.

Proposition 3.2. *For AV, PH, and EQ, $\text{cost-red}_E(p)$ can be computed alongside the rule, at an $O(1)$ cost per round.*

3.2. Add-Approvals Measures

Next we consider the number of additional approvals needed by a losing project to be funded. Our idea is to consider different attitudes toward risk when asking random voters to add approvals for a given project: An optimist would hope to be lucky and obtain new approvals exactly from the right voters, a pessimist would prefer to ensure that the project is funded irrespective of which voters add the approvals, and a middle-ground position is to require some fixed probability of success. Formally, we express this idea as follows.

Definition 3.3. *Let f be a PB rule, let $E = (P, V, B)$ be a PB instance, and let $p \in P \setminus f(E)$ be a losing project. Then:*

1. $\text{optimist-add}_E^f(p)$ is the smallest number ℓ such that it is possible to ensure that p is funded by choosing ℓ voters and extending their approval sets with p .
2. $\text{pessimist-add}_E^f(p)$ is the smallest number ℓ such that for each subset of ℓ voters who do not approve p , extending their approval sets with p ensures that p is funded.
3. $\text{50\%-add}_E^f(p)$ is the smallest number ℓ such that if ℓ voters selected uniformly at random (among those who originally do not approve p) extend their approval sets with p , then p is funded with probability at least 50%.

The optimist measure was previously considered by [Faliszewski et al. \[2017\]](#) in multiwinner voting, whereas the 50%-threshold one is inspired by an analogous notion used by [Boehmer et al. \[2021, 2022\]](#) in the single-winner setting, and by [Boehmer et al. \[2023\]](#) in the robustness analysis of PB outcomes. For each rule f , each PB instance E , and each losing project p we have the following:

$$\text{optimist-add}_E(p) \leq \text{50\%-add}_E(p) \leq \text{pessimist-add}_E(p).$$

For GREEDYAV all three measures are equal and immediate to compute, so we mostly focus on the proportional rules. Computing the optimist measure is easy as it suffices to consider each round independently and find the “richest” voters whose approval for p would lead to funding it.

Proposition 3.4. *For PH and EQ, $\text{optimist-add}_E(p)$ can be computed alongside the rule, at an $O(n \log n)$ cost per round, where n is the number of voters. For AV it can be computed alongside the rule at an $O(1)$ cost per round.*

On the negative side, deciding if the pessimist measure has at least a given value under PHRAGMÉN is coNP-complete (the result for EQUAL-SHARES remains elusive, although we also suspect computational hardness). Our proof looks at the complement of our problem, where we ask if it is possible to add a certain number of approvals for p without getting it funded, and we give a reduction from a variant of SET COVER where we ask for an exact cover (i.e., a family of pairwise disjoint sets that union up to a given universe). The idea is to set up a (somewhat intricate) PB instance where PHRAGMÉN interleaves between selecting candidates from the to-be-covered universe and certain dummy candidates, whose selection resets voters’ budgets to a fixed state. If we add approvals for a designated candidate p to voters that form an exact cover, then this process goes as planned, but if we add approvals to two voters whose corresponding sets overlap, then p gets funded.

Theorem 3.5. For PH, the problem of deciding if $\text{pessimist-add}_E(p)$ is at least a given value ℓ is coNP-complete, even if all projects have unit cost.

Fortunately, for PHRAGMÉN and EQUAL-SHARES we can compute the pessimist measure using an FPT algorithm parameterized by the number of originally funded projects. The key insight is to group voters by their bank account balances at the start of each round. Unlike many FPT algorithms, this one is indeed practical (since in practice the number of groups is small) and we use it in our experiments (using Gurobi).

Theorem 3.6. For PH and EQ, there is an algorithm that computes $\text{pessimist-add}_E(p)$ and runs in FPT time with respect to parameter $|f(E)|$, i.e., the number of rounds.

Proof sketch (PHRAGMÉN). Let $E = (P, V, B)$ be a PB instance with losing project p (for ease of exposition, we assume p to be last in the tie-breaking order). We will show how to compute the largest number of approvals whose addition *does not* lead to funding p ; $\text{pessimist-add}_E(p)$ is one larger. Let $k = |\text{PHRAGMÉN}(E)|$ be the number of funded projects or, equivalently, the number of rounds performed by the rule. We assume that p is approved by at least one voter in V .

For each round i , let m_i be the difference between $\text{cost}(p)$ and the total funds that approvers of p have in round i . Our goal is to find a largest group of voters who do not approve p and whose total funds in each round i are at most m_i .

For each voter $v_i \in V \setminus A(p)$, we define its *balance vector* $(b_1^i, \dots, b_k^i)$, which contains the balance of the voter's bank account right before each round. We partition the voters not approving p into *voter-types* $T = \{T_1, \dots, T_t\}$, where each type consists of voters with identical balance vectors.

To solve the problem, we form an integer linear program (ILP). For every voter-type T_i we have a nonnegative integer variable x_{T_i} that represents the number of voters of this type that will additionally approve p . For each round i we form the following *round-constraint*:

$$\sum_{j \in \{1, \dots, t\}} x_{T_j} \cdot b_i^{T_j} \leq m_i.$$

The objective function is to maximize the sum of all the x_{T_i} variables. Our algorithm outputs the value of this sum plus 1.

Finally, we note that there are at most $O(2^k)$ voter types (indeed, each voter type corresponds to a k -dimensional 0/1 vector, which has 1 in position i if a voter approves—and, hence, pays for—the candidate selected in round i). Thus, the number of variables in our ILP is $O(2^k)$; we can solve it using the classic algorithm of [Lenstra, Jr. \[1983\]](#) in FPT time with respect to k . The algorithm can be tweaked to also work for EQUAL-SHARES. $\square$

For the 50%-threshold measure, we resort to sampling. That is, given rule f , a PB instance E and a losing project p , we iterate over numbers ℓ of approvals to add and for each of them we repeat the following experiment t times (where t is a parameter): We add approvals for p to ℓ voters chosen uniformly at random (among those not approving p) and we compute f on the thus-modified instance. We terminate for the smallest value ℓ where p was funded at least $t/2$ times. We use sampling because the construction from Theorem 3.5 also shows that for PHRAGMÉN, the problem of evaluating the probability that a given project is funded after randomly adding a given number of approvals is #P-complete.

3.3. Add-Singletons Measure

Instead of asking existing voters to approve some project p , one can also recruit additional voters, who would only approve p . This gives rise to the following measure.

Definition 3.7. Let f be a PB rule, let $E = (P, V, B)$ be a PB instance, and let $p \in P \setminus f(E)$ be a losing project. Then $\text{singleton-add}_E^f(p)$ is the smallest number ℓ such that if we extend V with ℓ voters who only approve p , then f selects p .

For AV, this is equal to the measures from the previous section. For PHRAGMÉN, $\text{singleton-add}_E(p)$ is upper-bounded by $\text{optimist-add}_E(p)$: In each round the newly added voters always have at least as much money as the original ones (until p is selected). Under EQUAL-SHARES adding new voters changes the initial balances of the voters, which can change the overall execution of the rule and, hence, there is no clear relation between the two measures. In fact, under EQUAL-SHARES it is even possible that a project is funded after adding ℓ voters, but may fail to be funded after adding $\ell + 1$ of them [Lackner and Skowron, 2023, Proposition A.3]; see also the experiments in Section 4.1.

Computing the value of $\text{singleton-add}_E(p)$ is easy for PHRAGMÉN because we can compute how much additional funds each new voter would bring in each round. For EQUAL-SHARES, due to its nonmonotonic behavior, we resort to a brute-force approach: We keep adding voters one by one and recompute the rule each time (in our experiments, we were forced to add larger groups to speed up computation).

Proposition 3.8. For AV and PH, $\text{singleton-add}_E(p)$ can be computed alongside the rule, at an $O(1)$ cost per round.

3.4. Rivalry-Reduction Measure

Under proportional rules, a project may lose because its supporters also approve other projects, on which they spend their virtual money. Thus, another strategy that a project proposer could employ to increase a project's chances of success is to try to convince its supporters to not approve other projects.

Definition 3.9. Let f be a PB rule, let $E = (P, V, B)$ be a PB instance, and let $p \in P \setminus f(E)$ be a losing project. Then $\text{rival-red}_E^f(p)$ is the smallest number ℓ such that if we select ℓ voters uniformly at random (among those who approve p) and change them to only approve p , then p is funded with probability at least 50%.

This measure is not always defined: A project with too few voters will not win even if they do not support any other projects. For the sake of focus, we do not study the optimist and pessimist variants, except to note that even the optimist variant would be NP-complete to compute (by adapting proofs on bribery and control in single-winner approval voting [Faliszewski et al., 2009, Hemaspaandra et al., 2007]).

Proposition 3.10. For AV, PH, and EQ, the problem of deciding, given a PB instance E , a losing project p , and an integer ℓ , if it is possible to ensure p 's victory by changing at most ℓ votes that originally approve p to only approve p is NP-complete, even when $B = 1$ and every project costs 1.

To compute rival-red, we use an analogous sampling approach as in the case of 50%-add.

One may also wonder why we consider *all* the voters who approve p and not only those who approve p and some further project(s). We chose this approach to capture a campaign that reaches the supporters of the project randomly.

4. The Measures in Practice

Next, we analyze the behavior and correlation of our measures on real-world PB instances from Pabulib Faliszewski et al. [2023], focusing on EQ/PHRAGMÉN here. Afterwards, we present a

	optimist	pessimist	50%	singleton	rival	cost
optimist	—	0.87	0.98	0.98	0.88	0.76
pessimist	0.87	—	0.94	0.84	0.50	0.73
50%	0.98	0.94	—	0.95	0.78	0.79
singleton	0.98	0.84	0.95	—	0.93	0.74
rival	0.88	0.50	0.78	0.93	—	0.63
cost	0.76	0.73	0.79	0.74	0.63	—

Table 2: Pearson Correlation Coefficients between measures for EQ/PHRAGMÉN (values near 1 mean strong correlation).

detailed study of the PB election from Wieliczka. We include further details, as well as results for PHRAGMÉN and GREEDYAV, in [Appendix C](#).

Data. We conduct our experiments on all 551 PB instances with approval votes from Pabulib [Faliszewski et al. \[2023\]](#) for which both PHRAGMÉN and EQ/PHRAGMÉN terminate within one second (on 1 thread of an Intel(R) Xeon(R) Gold 6338 CPU @ 2.00GHz core). In total, there are 3639 losing projects for EQ/PHRAGMÉN and 3513 for PHRAGMÉN.

Measures. To compute the measures, we use the algorithms described in [Section 3](#). For the brute-force and sampling algorithms, we increase the approval score of the designated project by 1% in each step (repeating each step 100 times for the sampling algorithms). To simplify comparisons between measures, we normalize them to lie between 0 (being far away from victory) and 1 (being close to victory). Specifically, for our four measures modifying a project’s approval score, we divide its original approval score by its approval score plus the measure. For example, if a project with score 20 requires 80 additional approvals (according to the measure), then the normalized value is 0.2, since the project received 20% of the needed approvals. For cost-red, we divide $\text{cost-red}(p)$ by $\text{cost}(p)$, and for rival-red, we report the fraction of supporters who can continue to approve other projects.

Running Times. We ran our experiments on 10 threads of an Intel(R) Xeon(R) Gold 6338 CPU @ 2.00GHz core. The algorithms for optimist-add, pessimist-add, singleton-add, and cost-red are all very fast, finishing in below 20 seconds on 95% of instances. The sampling-based algorithms for rival-red and 50%-add are naturally slower, but still finish in 88% of cases in below 10 minutes. In sum, while our implementations are certainly not fully optimized, running the different algorithms for a PB exercise in one’s city is feasible, even for tens of thousands of voters and hundreds of projects.

4.1. Behavior and Correlation

In [Table 2](#), we show the linear correlation between our measures for EQ/PHRAGMÉN, as given by the Pearson Correlation Coefficient (PCC) and in [Appendix C.1](#), we show correlation plots. In general, we want to stress that we observe many projects close to getting funded under the different measures (such projects can also be found in some of the plots presented in this section).

Add-Approvals and Singletons Measure. We first analyze the four measures related to adding approvals to existing or new ballots. As seen in [Table 2](#), they all have a strong pairwise correlation. However, there are small differences motivating a partitioning of the measures into two groups: optimist-add, 50%-add, and singleton-add all have a pairwise correlation of at least 0.95, whereas

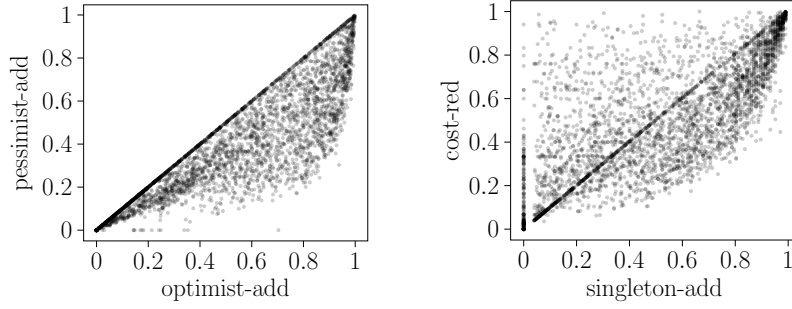


Figure 1: Correlation plots where each point is one project. Measures are normalized so that 1 denotes no change and 0 denotes a maximal-size change.

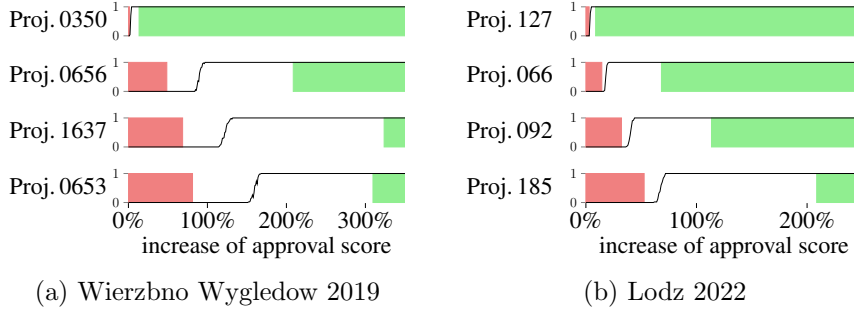


Figure 2: Line plots showing how the funding probability of a project develops from 0 to 1 when increasing its approval score by adding approvals uniformly at random to existing voters. The red area goes until the optimist-add value and the green area extends from the pessimist-add value.

pessimist-add has a lower correlation with the other three measures. In the first group, the correlation between optimist-add and singleton-add is particularly strong. For more than 90% of the projects, the difference between the two measures is less than 0.051 (exceptions include projects where one of the two measures is zero).

Comparing the optimistic and pessimistic views on adding approvals, while they have a strong correlation of 0.87, on the level of single projects, they can produce quite different results (see Figure 1 left). In fact, for around 10% of projects, almost twice as many approvals are needed to get the project funded under the pessimistic approach than under the optimistic one. Thus, under EQ/PHRAGMÉN, it really matters which voters add approvals for a project.

The 50%-add measure lies between optimist-add and pessimist-add and is strongly correlated with them. It tends to be slightly closer to the optimistic view (average difference 0.063) than to the pessimistic one (0.077).

In Figure 2, we show for two instances how the funding probabilities of projects evolve when adding approvals uniformly at random to existing voters. Projects very quickly transition from having a funding probability close to 0% to having one close to 100%, even for projects where there is a large gap between optimist-add and pessimist-add. This phase-transition-like behavior appears for almost all projects. Thus, for practical purposes, reasonably optimistic and pessimistic views coincide (i.e., requiring that a project gets funded with at least 2% probability is very similar to requiring that it gets funded with at least 98% probability). Thus, 50%-add gives us a good estimate of how many approvals a project “practically misses” to get funded.

If one wants to reduce the number of measures, it is probably simplest to use singleton-add, as it is strongly correlated with optimist-add and 50%-add and can be explained very easily. Another advantage is that even for more complicated or slower rules, where efficient algorithms

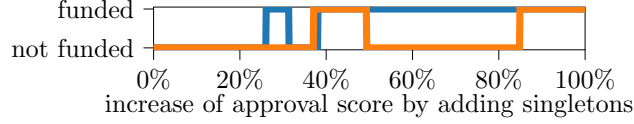


Figure 3: Behavior of two projects when adding voters who only support the project, taken from Warsaw 2023 (Praga-Polnoc, in blue) and Warsaw 2017 (Goclaw, in orange).

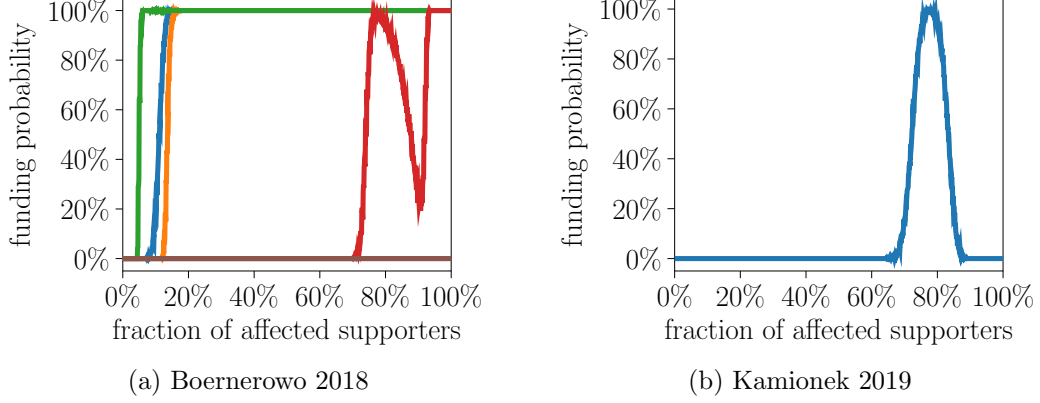
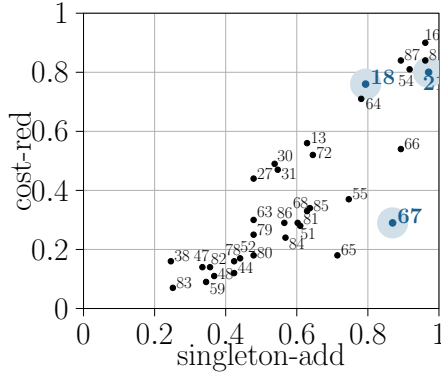


Figure 4: Line plots showing how the funding probability of a project develops if we remove rivalry approvals from its supporters selected uniformly at random (each line corresponds to a single project; all non-funded projects are shown).

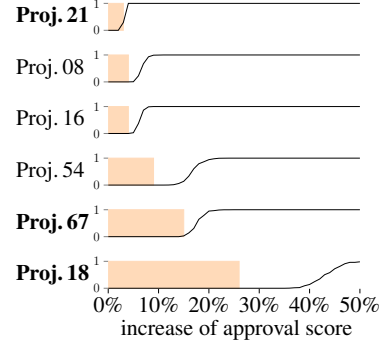
for the other measures might be hard to find, singleton-add can be computed via brute-force (this approach is unsatisfying algorithmically, but if one is simply to provide information to participants of a PB election once a year, then it typically suffices). The only drawback of this measure for EQUAL-SHARES is that, as discussed in [Section 3.3](#), adding singletons can make a project lose. In fact, this occasionally happens on our data in unexpected ways (see [Figure 3](#) for examples).

Rivalry-Reduction Measure. Our sampling-based algorithm for rival-red returns a value for 1605 out of the 3639 not-funded projects, which in particular implies that we can get these projects funded after removing all other approvals from the ballots of some (or all) of their supporters. **This highlights the general power of lobbying one’s supporters to only approve a single project in an election using a proportional rule.** Regarding the correlation of rival-red with the other measures, [Table 2](#) shows the PCC correlation on the 1605 projects where rival-red returns a value. The correlation with the adding approval measures is strong (but not very strong). One interesting observation is that for projects where optimist-add is below 0.5 (i.e., for projects whose approval score needs to at least double to be able to win), removing rivalry approvals is almost never sufficient to get funded. Regarding the 2034 projects that cannot be funded after removing rivalry approvals, they have a very diverse performance with regard to the other measures. **For example, removing rivalry approvals might not be sufficient even for projects that are only missing a few approvals.**

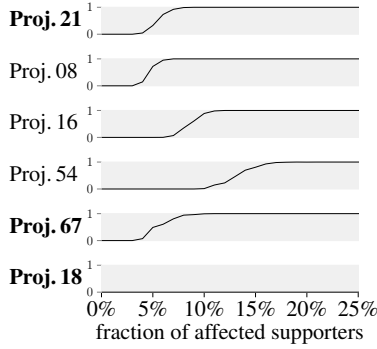
Interestingly, removing more rivalry approvals does not necessarily help a project. The reason for this is that by removing the approval of a rival, we can modify the execution of EQ/PHRAGMÉN in arbitrary rounds and thereby also help other projects to get funded. **Interestingly, this non-monotonic behavior of projects appears in many forms.** [Figure 4](#) shows two examples. Nevertheless, for rivalry reduction, for most projects there is also a quick jump from an almost 0% to an almost 99% funding probability.



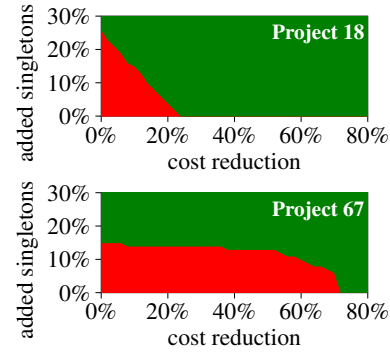
(a) Correlation plots for cost-red and singleton-add. Larger font and pale background indicate the three projects that we focus on.



(b) Funding probability when adding approvals uniformly at random to voters. Orange areas go to the singleton-add value.



(c) Funding probability when removing rivalry approvals from some uniformly at random selected group of supporters.



(d) Tradeoff between fractional reduction of costs and increase of approval score by adding singletons.

Figure 5: Information on the Wieliczka 2023 Green Million election.

Cost-Reduction Measure. The cost reduction measure is less correlated to the other measures, which is expected since it is the only measure that modifies project costs. Yet, as all our measures help projects in some way, it is intuitive that there is a certain correlation (of around 0.75) to the other measures. To analyze the tradeoff between adding approvals and reducing costs in more detail, Figure 1 (right) shows the connection between cost-red and singleton-add. We see that for a majority of projects, adding approvals is more powerful than reducing the project’s cost. However, there are also numerous projects where it is the other way around, which makes it hard to make a general recommendation.

4.2. Wieliczka’s Green Million

We conclude by constructing an information package for the PB election held in Wieliczka in 2023, where EQUAL-SHARES was used for the first time (<https://equalshares.net/resources/zielony-milion/>). This PB election focused on ecological issues, with 64 projects placed on the ballot. 6586 people cast their votes. Each project cost up to 100 000 PLN, and each voter could approve any number of projects. The budget was one million PLN (approximately 225 000 EUR). The vote data is available on Pabulib. Notably, Green Million used a more involved completion method of EQUAL-SHARES which increases the initial balances of the voter’s bank accounts in small steps (see Appendix A for a description). This is computationally intensive, so we focus on

the easier-to-compute measures: cost-red, singleton-add, rival-red, and 50%-add.

Figure 5 shows the results. In particular, Figure 5 (a) depicts the correlation between singleton-add and cost-red, Figure 5 (b) how the funding probabilities of projects change when adding approvals to existing voters at random, and Figure 5 (c) their behavior when removing rivalry approvals. Below, for three projects cherry-picked for interesting conclusions, we discuss the contents of a hypothetical information package produced for the Green Million PB, and what project proposers could learn from it. The analysis reinforces the conclusion that projects close to being funded regularly appear and that our measures contribute different perspectives.

Project 21 (cost 100 000 PLN, 496 votes). According to our measures, this project was very close to winning. Indeed, it needs only about 3% additional singleton votes or additional approvals to be funded. It was also close to winning in terms of rivalry reduction, in the sense that relatively few of its supporters (below 10%) would need to refrain from supporting other projects in order for Project 21 to be funded. On the other hand, the cost-reduction measure shows that the project would have to be 20% cheaper to be selected, given the current votes. This indicates a strong project that probably should be resubmitted in the next edition of the program, with a slightly more aggressive support campaign. Reducing the cost of the project would be somewhat less effective.

Project 18 (cost 51 000 PLN, 163 votes). The project performs similarly under singleton-add and cost-red; both measures indicate that the project was about 80% on the way to winning. With respect to both of these measures, only 5 other losing projects do better. However, Project 18's 50%-add performance is much worse, and the project continues to lose even if we remove rivalry approvals from all its supporters. This indicates that the main problem of Project 18 is insufficient votes: Even in the first round of EQUAL-SHARES, its supporters don't have enough money available to buy the project. Adding approvals to existing voters helps less with this funding gap, as these voters might spend their budget in earlier rounds on more popular projects.

Project 67 (cost 16 500 PLN, 140 votes). Project 67 needs around 15% more approvals according to singleton-add and 50%-add, but its cost would need to be reduced by 70% to get funded. Thus Project 67 is not so far off in terms of the number of supporters it has, but under EQUAL-SHARES, those supporters spend almost all of their money on more popular projects before Project 67 is considered by the rule. This interpretation is confirmed by its performance with respect to rival-red, where its performance matches projects like Project 21, which is very close to being funded according to all measures. Hence, Project 67's main issue is competition with projects supported by the same voters.

Combined Strategy. One might also ask whether a combined strategy of slightly lowering the cost of a project and getting a few more singleton voters could be effective for our projects. We illustrate this approach in Figure 5 (d). The x -axis shows the percentage by which the cost is reduced, and the y -axis shows the increase of the approval score of the project (by adding singletons). A point is colored green if the project would be funded if both actions were performed, and red if it would continue to lose. We see that for Project 21 we can exchange additional votes for cost reduction in a linear way, but for Project 67 the behavior is highly nonlinear.

5. Future Work

We have begun the work of designing useful information packages for voters and project proposers. In future work, it would be useful to collect feedback from participants about what measures

and visualizations are of most interest. There is also room for developing additional measures or improving some of the presented ones. For instance, as an additional measure, one could consider changes in the overall budget, to be more precise, the minimum budget that is needed for the designated project to get funded. Moreover, for presented measures that increased the score of project p , one could try to prioritize adding approvals to voters who are most likely to vote for p (e.g., because they are similar to current supporters of p).

Acknowledgments

The main work was done while Niclas Boehmer was affiliated with TU Berlin, where he was supported by the DFG project ComSoc-MPMS (NI 369/22). This work was co-funded by the European Union under the project Robotics and advanced industrial production (reg. no. CZ.02.01.01/00/22_008/0004590). Šimon Schierreich also acknowledges the support of the Grant Agency of the Czech Technical University in Prague, grant No. SGS23/205/OHK3/3T/18. This project has received funding from the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation programme (grant agreement No 101002854). Piotr Skowron was supported by the European Research Council (ERC; project PRO-DEMOCRATIC, grant agreement No 101076570).



6. Bibliography

- Dorothea Baumeister, Linus Boes, and Johanna Hillebrand. Complexity of manipulative interference in participatory budgeting. In *Proceedings of ADT-2021*, pages 424–439. Springer, 2021. [→ p. 5, 6]
- N. Boehmer, R. Brederbeck, P. Faliszewski, and R. Niedermeier. Winner robustness via swap- and shift-bribery: Parameterized counting complexity and experiments. In *Proceedings of IJCAI-2021*, pages 52–58, 2021. [→ p. 6]
- N. Boehmer, R. Brederbeck, P. Faliszewski, and R. Niedermeier. A quantitative and qualitative analysis of the robustness of (real-world) election winners. In *Proceedings of EAAMO-2022*, pages 7:1–7:10, 2022. [→ p. 6]
- N. Boehmer, P. Faliszewski, L. Janeczko, and A. Kaczmarczyk. Robustness of participatory budgeting outcomes: Complexity and experiments. In *Proceedings of SAGT-2023*, 2023. [→ p. 6]
- M. Brill, R. Freeman, S. Janson, and M. Lackner. Phragmén’s voting methods and justified representation. In *Proceedings of AAAI-2017*, pages 406–413, 2017. [→ p. 2, 4]
- Markus Brill, Stefan Forster, Martin Lackner, Jan Maly, and Jannik Peters. Proportionality in approval-based participatory budgeting. In *Proceedings of AAAI-2023*, pages 398–404, 2022. [→ p. 4]
- Y. Cabannes. Participatory budgeting: A significant contribution to participatory democracy. *Environment and Urbanization*, 16(1):27–46, 2004. [→ p. 2]
- David Cary. Estimating the margin of victory for instant-runoff voting. 2011. Presented at 2011 Electronic Voting Technology Workshop/Workshop on Trustworthy Elections. [→ p. 5]

- R. Fairstein, G. Benadè, and K. Gal. Participatory budgeting designs for the real world. In *Proceedings of AAAI-2023*, pages 5633–5640, 2023. [→ p. 2]
- P. Faliszewski and J. Rothe. Control and bribery in voting. In F. Brandt, V. Conitzer, U. Endriss, J. Lang, and A. D. Procaccia, editors, *Handbook of Computational Social Choice*, chapter 7. Cambridge University Press, 2015. [→ p. 5]
- P. Faliszewski, E. Hemaspaandra, and L. Hemaspaandra. How hard is bribery in elections? *Journal of Artificial Intelligence Research*, 35:485–532, 2009. [→ p. 5, 8]
- P. Faliszewski, P. Skowron, and N. Talmon. Bribery as a measure of candidate success: Complexity results for approval-based multiwinner rules. In *Proceedings of AAMAS-17*, pages 6–14, 2017. [→ p. 5, 6]
- Piotr Faliszewski, Jarosław Flis, Dominik Peters, Grzegorz Pierczyński, Piotr Skowron, Dariusz Stolicki, Stanisław Szufa, and Nimrod Talmon. Participatory budgeting: Data, tools and analysis. In *Proceedings of IJCAI-2023*, pages 2667–2674, 8 2023. [→ p. 3, 5, 8, 9]
- A. Goel, A. Krishnaswamy, S. Sakshuwong, and T. Aitamurto. Knapsack voting for participatory budgeting. *ACM Transactions on Economics and Computation*, 7(2):1–27, 2019. [→ p. 2]
- T. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985. [→ p. 19]
- E. Hemaspaandra, L. Hemaspaandra, and J. Rothe. Anyone but him: The complexity of precluding an alternative. *Artificial Intelligence*, 171(5–6):255–285, 2007. [→ p. 8]
- Richard M. Karp. Reducibility among combinatorial problems. In Raymond E. Miller, James W. Thatcher, and Jean D. Bohlinger, editors, *Complexity of Computer Computations*, pages 85–103, Boston, MA, 1972. Springer. ISBN 978-1-4684-2001-2. doi: 10.1007/978-1-4684-2001-2_9. [→ p. 26]
- M. Lackner and P. Skowron. *Multi-Winner Voting with Approval Preferences*. Springer Briefs in Intelligent Systems. Springer, 2023. [→ p. 8]
- H. Lenstra, Jr. Integer programming with a fixed number of variables. *Mathematics of Operations Research*, 8(4):538–548, 1983. [→ p. 7, 24]
- M. Los, Z. Christoff, and D. Grossi. Proportional budget allocations: Towards a systematization. In *Proceedings of IJCAI-2022*, pages 398–404, 2022. [→ p. 2, 4]
- T. Magrino, R. Rivest, E. Shen, and D. Wagner. Computing the margin of victory in IRV elections. Presented at 2011 Electronic Voting Technology Workshop/Workshop on Trustworthy Elections, August 2011. [→ p. 5]
- D. Peters and P. Skowron. Proportionality and the limits of welfarism. In *Proceedings of EC-20*, pages 793–794, 2020. [→ p. 2, 4]
- D. Peters, G. Pierczynski, and P. Skowron. Proportional participatory budgeting with additive utilities. In *Proceedings of NeurIPS-2021*, pages 12726–12737, 2021. [→ p. 2, 4]
- S. Rey and J. Maly. The (computational) social choice take on indivisible participatory budgeting. Technical Report arXiv.2303.00621, arXiv, 2023. [→ p. 2]
- L. Xia. Computing the margin of victory for various voting rules. In *Proceedings of EC-12*, pages 982–999, June 2012. [→ p. 5]

Y. Yang. On the complexity of destructive bribery in approval-based multi-winner voting. In *Proceedings of AAMAS-2020*, pages 1584–1592, 2020. [[→ p. 5](#)]

A. Additional Material for Section 2

In this section, we describe the variant of EQUAL-SHARES used in Wieliczka's Green Million election in 2023.

In the first round, the standard EQUAL-SHARES procedure is run. If the elected outcome is not exhaustive, the execution of EQUAL-SHARES is repeated with incremented values of voters' initial endowments (so that they are equal to $B/|V| + 1$, $B/|V| + 2$, ... in further rounds). The procedure stops after the i th round if one of the following conditions is satisfied:

1. An exhaustive (with respect to the budget B) outcome W has been elected. Then W is returned.
2. The total cost of the elected outcome exceeds B . Then the outcome W elected after the $(i - 1)$ th round is returned. Note that in this case, we know that $i > 1$ (since in the first round there are only B money units in the system and the cost of every elected project is covered) and the cost of W does not exceed B .

At each step of the procedure, possible ties between projects are broken according to a randomly chosen priority ranking of projects.

B. Additional Material for Section 3

B.1. When Are Our Measures Undefined

Below we list under what conditions are our measures defined:

Cost reduction. For each of our rules, the cost-reduction measure is well-defined for projects that received at least a single approval (in the worst case we lower the cost to zero; for GREEDYAV even the single approval is not necessary).

Adding approvals. The three adding-approvals measures (optimist, pessimist, and the randomized one) are defined in almost all cases: It suffices that all the voters approve a given project p . The exception happens if there is another project that already is approved by all the voters, the rule selects it prior to p (e.g., due to tie-breaking or the cost), and the remaining budget is insufficient for p .

Adding singletons. One can verify that the adding-singletons measure is well-defined for GREEDYAV and PHRAGMÉN. For EQUAL-SHARES, it is well-defined provided that project p that we consider costs less than the available budget or it costs as much as the budget and is supported by all the voters.

B.2. Cost-Reduction Measure

Proposition 3.2. *For AV, PH, and EQ, $\text{cost-red}_E(p)$ can be computed alongside the rule, at an $O(1)$ cost per round.*

Proof. Recall that the AV rule considers projects in the order of their non-increasing approval scores (that is, the order depends solely on the number of approvals and is not affected by the cost of the projects) and a project is included in the outcome W if doing so will not exceed the available budget. Let $i \in \mathbb{N}$ be the round in which the rule considers project p and B' be the remaining budget at the beginning of round i . Since p was not initially funded, we have $\text{cost}(p) > B'$, however, setting $\text{cost}(p) = B'$ makes the project affordable. Therefore, $\text{cost-red}_E^{\text{AV}}(p) = B'$.

For PH, we first ignore the project p and execute the rule with the remaining candidates. This rule in r time steps adds some projects to the outcome. For each time step $t_1, \dots, t_r$,

we compute the endowments $E_1, \dots, E_r$ of supporters of our distinguished candidate p . The result is then the maximum x_i on all time steps $i \in [r]$ in which $\text{cost}(W) + x_i \leq B$, where $x_i = \min\{E_i, B - \text{cost}(W)\}$. Suppose that x is the maximum possible cost so that even after setting $\text{cost}(p) = x$, the project p is funded. According to the definition of the rule, the project is funded in some time step t_i , $i \in [r]$. If $x > B - \text{cost}(W)$, then p cannot be added to the outcome since there is not enough remaining budget. If $x > E_i$, then p cannot be founded in this round, as the endowments of supporters of p are not high enough. Therefore, x is at most $\min_{j \in [r]} \{E_j, B - \text{cost}(W)\}$, which is exactly the result of our algorithm.

Finally, for EQ, we again examine the execution of the rule without the project p . We compute the value for each iteration separately and then return the maximum. Assume that in round i project d is funded as it is q -affordable. Then we want to find a cost value x such that $\sum_{v_j \in A(c)} \min(b_i(v_j), q \cdot x) = x$. Observe that $\sum_{v_j \in A(c)} \min(b_i(v_j), q \cdot x)$ is monotonically increasing and piecewise linear. Let $b^1 \leq \dots \leq b^n$ be the budget values of the voters. Then for $x \in \left[\frac{b_i}{q}, \frac{b_{i+1}}{q}\right]$ the function is linear as $(n - i) \cdot q \cdot x + \sum_{j=1}^i b_j$. We can first find the value of b_i such that the solution lies within $\left[\frac{b_i}{q}, \frac{b_{i+1}}{q}\right]$ and then simply check within this interval. If the solution lies in no such interval, then we know $x > b^n$ or $x < b^1$. We can handle these cases. $\square$

B.3. Add-Approvals Measure

Optimistic Add

Proposition 3.4. *For PH and EQ, $\text{optimist-add}_E(p)$ can be computed alongside the rule, at an $O(n \log n)$ cost per round, where n is the number of voters. For AV it can be computed alongside the rule at an $O(1)$ cost per round.*

Proof. We start with the AV rule. Crucial observation in this case is that for the rule, it does not matter where we add approvals. So, we run the rule on the original instance as usual. The interesting part of the process takes place once the funding of the currently processed project c first decreases the value of the remaining budget below $\text{cost}(p)$. Before funding c , we compute m^* as $|A(c)| - |A(p)| + [c \succ p]$, where $[c \succ p]$ evaluates to 1 if c precedes p in tie-breaking order and to 0 otherwise. If $m^* + |A(p)| > |V|$, then the measure is not defined, and we terminate. Otherwise, the algorithm outputs the value of m^* . For the correctness, suppose that $\text{optimist-add}_E^{\text{AV}}(p) = m' < m^*$. If we add m' additional voters that approve p , then p is assumed by the rule in some round after c is funded. However, c was selected so that, after funding c , the remaining budget is below $\text{cost}(p)$. Consequently, adding m' additional voters is not sufficient to fund p and therefore $\text{optimist-add}_E^{\text{AV}}(p) \geq m^*$. Since, by the definition of the rule, the distinguished project p is funded after adding m^* approvals, the algorithm is correct and clearly performs only $O(1)$ additional operations per round.

For PH, the high-level idea is to examine each round separately and return the minimum number of additional approvals needed to fund p . More formally, let i be a round such that there is still enough money to fund p (otherwise, we stop the execution of the rule and return the currently stored minimum). Let f_i be the money that the distinguished candidate lacks in order to be funded in round i , and let $X \subseteq V$ be the set of voters not approving p . We sort the voters in X according to their current balance and select the minimum possible number m_i of voters (in descending order according to their balances) such that the current endowment of p plus the summed endowments of selected voters from X is at least $\text{cost}(p)$. The outcome of the algorithm is then the minimum for all m_i . In each round, we additionally sort the voters in X according to their current balance, which can be done in $O(n \log n)$ time, where $n = |V|$.

Finally, with EQ we look again at each round where there is still enough money left to buy p , compute the number of additional approvals we need to fund p in this round, and return the

minimum over these values for all the examined rounds. Let i be a round currently processed, q_i be the minimum q such that a project is q -affordable in round i , and f_i be the missing money to fund p . Additionally, we set $X \subseteq V$ as the set of voters that do not approve p and e_v^i as the endowment of the voter $v \in X$ in the round i . We sort the voters in X in descending order according to their endowments; assume that $e_{x_1}^i \geq \dots \geq e_{x_n}^i$. Now, we find the minimum m_i such that $\sum_t^{m_i} \min(e_{x_t}^i, \text{cost}(p) \cdot q_i) \geq \text{cost}(p)$. If there are still enough funds in the final round, we do the same thing with $q_i = \infty$. Is in the case of PH, in every round we additionally sort all voters in X and do a linear-time computation to determine m_i . That is, for EQ, we again need $O(n \log n)$ additional operations per round. $\square$

Pessimist Add

Theorem 3.5. *For PH, the problem of deciding if $\text{pessimist-add}_E(p)$ is at least a given value ℓ is coNP-complete, even if all projects have unit cost.*

Proof. Membership in coNP is clear: Given an instance of a problem it suffices to guess ℓ voters and add the designated project to their approval sets. Then, accept on computation paths where the project is funded and reject on those where it is not. If $\text{pessimist-add}_E^{\text{PH}}(p) \leq \ell$, then all paths accept and otherwise at least one path rejects.

To show coNP-hardness, we will work with a complement of our problem: Given a PB instance E , a losing project p , and nonnegative integer ℓ , we ask if it is possible to add ℓ approvals for p without getting it to be funded. We show that this problem is NP-hard by giving a reduction from RX3C, a restricted variant of the EXACT COVER BY 3-SETS problem [Gonzalez \[1985\]](#). In this problem we are given a universe $U = \{u_1, u_2, \dots, u_n\}$ of elements and a collection $\mathcal{S} = \{S_1, S_2, \dots, S_n\}$ of size-3 subsets of U . Each element $u_i \in U$ appears in exactly three sets from $\mathcal{S}$ and n is divisible by three. We ask whether there exists an exact cover over U , that is, a collection of sets from $\mathcal{S}$ such that each element from U appears in exactly one set. In this case, since each set has size three, each valid exact cover must have size $n/3$.

W.l.o.g., we assume that $n \geq 60$ and $n = 54q + 6$ for some $q \in \mathbb{N}$. Otherwise, we could keep adding three new elements e_1, e_2, e_3 and three new (identical) sets $\{e_1, e_2, e_3\}, \{e_1, e_2, e_3\}, \{e_1, e_2, e_3\}$ until our condition were met (we interpret $\mathcal{S}$ is a multiset). These newly added sets do not interfere with the original ones, so the solution for the modified instance always consists of a solution for the original one plus one copy of each newly created set.

Construction. We form a PB instance with project set $P = \{p\} \cup U \cup D \cup \{b_1, b_2\}$, where p is the designated project, U is the universe (so each of its members doubles as a project), $D = \{d_1, \dots, d_n\}$ is a set of dummy candidates, and projects b_1 and b_2 are used to initiate appropriate bank-account balances for the voters. Each project has the same unit cost. To introduce the voters, we need the following notation:

$$\begin{aligned} w &= 1/18(11n^{10} + 3n^9), \\ z &= 1/18(5n^{10} + 3n^9), \\ y &= 1/18(4n^{10} - 3n^9), \\ \alpha &= 1/27(4n^{10} - 3n^9 - 6n - 18), \text{ and} \\ s &= n^9/6. \end{aligned}$$

Numbers z, y, w and s are positive integers because n is divisible by 6. Further:

$$\begin{aligned} \alpha &= 1/27(4n^{10} - 3n^9 - 6 \cdot (54q + 6) - 18) \\ &= 1/27(4n^{10} - 3n^9 - 12 \cdot 27q - 2 \cdot 27), \end{aligned}$$

so α also is a positive integer. Finally, we see that $\alpha < y$. We create four groups of voters as follows:

Group W : We have w voters approving only b_1 and b_2 .

Group Z : We have z voters approving candidates $\{p\} \cup U$.

Group Y : We have y voters approving candidates $D \cup \{b_1, b_2\}$. Further, first α of them also approve p (we refer to them as group A ; naturally, the A -voters also belong to group Y).

Group S (Set Voters): For each $S_i \in \mathcal{S}$, we create s voters approving candidates $D \cup \{b_1, b_2\} \cup (U \setminus S_i)$. That is, each of them approves $n + 2 + (n - 3) = 2n - 1$ candidates (recall that $|S_i| = 3$). There are $s \cdot n$ set voters in total and each universe candidate is approved by exactly $s \cdot (n - 3)$ of them.

Finally, we set the budget to be $B = 2n + 2$, and we set the tie-breaking order to be:

$$b_1 \succ b_2 \succ p \succ u_1 \succ d_1 \succ u_2 \succ d_2 \succ \dots \succ u_n \succ d_n.$$

We ask if we can add $\ell = n/3$ approvals in such a way that p does not get funded. We introduce the following constants:

$$t_1 = 1/n^{10}, \quad t_2 = 2/n^{10},$$

representing time periods after which we buy some candidates (this will become clear a bit later).

Intuitive Idea. Let us now explain the intuition behind our construction. For the sake of simplicity, let us assume that approvals can only be added to the set voters. Later we will show why this assumption is valid.

Consider our PB instance after ℓ extra approvals for p were added to some of the set voters. PHRAGMÉN proceeds as follows: First, at time $t_2/2$ the rule selects b_1 and at time t_2 the rule selects b_2 . Consequently, at this moment each of the Z -voters has budget t_2 and all other voters have budget 0. Then, there are n iterations which consist of selecting members of U and D in an interleaved fashion. At the beginning of the i -th iteration, each Z -voter has budget t_2 , whereas the set voters and the Y -voters have budgets 0. The W -voters do not play any role since all their projects have already been purchased. After time t_1 (within the iteration) project u_i is purchased by the voters from group Z and the set voters corresponding to sets that do not include u_i . Project p cannot be purchased at this moment regardless of how we add the approvals. Then, after time t_2 since purchasing u_i , the voters either buy d_i or p (the latter happens if we added approvals to at least two set voters that did not pay for u_i , i.e., to two set voters who correspond to sets that include u_i). If the voters purchase d_i instead of p , then all the Y -voters and all the set voters have their budgets reset to 0, whereas each Z -voter has budget t_2 . Hence the situation is the same as at the beginning of the iteration. Then the $(i + 1)$ -th iteration starts.

After n iterations we run out of budget. If p were not selected, then it means that for each iteration, after buying a universe project we purchased a project from D . This means that we must have added approvals for p to set voters that correspond to an exact cover of U (the reverse direction is also immediate: If we add approvals to set voters corresponding to an exact cover then p is not purchased). In other words, p can lose after adding $\ell = n/3$ approvals if and only if there exists an exact cover in our RX3C instance.

Formal Argument (Approvals Added to Set Voters Only). Now we formally argue that the execution of the rule proceeds as described above, assuming that all the additional approvals for p went to the set voters (below we give a number of (in)equalities; we give the somewhat tedious explanation as to why they hold at the end of the proof):

Initialization. Altogether, there are $y + s \cdot n + w$ voters who approve b_1 (and b_2). Thus, after time $t_2/2$ since the beginning of the execution of PHRAGMÉN, voters supporting b_1 have enough funds to purchase it (and, analogously, after the following $t_2/2$ chunk of time they have enough funds to also purchase b_2). Indeed, we have that:

$$(y + s \cdot n + w) \cdot t_2/2 = 1.$$

All the other projects have far fewer approvals, so b_1 and b_2 are purchased first. Consequently, at time t_2 since the beginning of the execution, b_1 and b_2 are funded.

Budgets at the beginning of each iteration. Assuming that project p has not been funded, at the beginning of each iteration each Z -voter has budget t_2 and all the other voters have budgets 0. This is exactly the situation right after project b_2 is selected, when the first iteration starts.

Iteration i until time t_1 . Let us consider what happens during the i -th iteration until time t_1 since its beginning. First, as witnessed by the following equality, at time t_1 voters supporting u_i have enough funds to purchase it:

$$z \cdot (t_1 + t_2) + s \cdot (n - 3) \cdot t_1 = 1.$$

(Indeed, each Z -voter had t_2 money at the beginning of the round and obtained another t_1 until time t_1 ; further, each of the $s \cdot (n - 3)$ set voters that approve u_i had budget 0 at the beginning of the round and earned t_1 until this moment.) Naturally, voters supporting projects $u_{i+1}, \dots, u_n$ also have sufficient amounts of money to buy them, but they lose to u_i due to the tie-breaking order (projects $u_1, \dots, u_{i-1}$ were purchased in prior iterations³). It remains to argue that neither p nor any of the D -projects can be purchased at time t_1 or prior to it. For the case of p , we see that at time t_1 its voters have total budget:

$$z \cdot (t_1 + t_2) + \ell \cdot t_1 + \alpha \cdot t_1 < 1$$

which does not suffice to buy p . Finally, at time t_1 since the beginning of the iteration, each D -project is supported by voters who in total have budget:

$$y \cdot t_1 + s \cdot n \cdot t_1 < 1$$

and, so, neither of the D -projects can be purchased. Consequently, at time t_1 since the beginning of the iteration, project u_i is funded and its supporters have their bank accounts reset to 0.

Iteration i from time t_1 until time $t_1 + t_2$. After time $t_1 + t_2$ since the beginning of the iteration, voters approving project d_i have enough funds to purchase it. Indeed, we have:

$$y \cdot (t_1 + t_2) + s \cdot n \cdot t_2 + 3s \cdot t_1 = 1$$

(note that by this time the Y -voters earned $t_1 + t_2$ each, each set voter earned t_2 since time t_1 , and the budgets of the $3s$ set voters corresponding to sets that include u_i were not

³Indeed, formally we should express our reasoning as an inductive proof, but we believe that the current approach is sufficiently clear and a bit lighter.

reset to zero at time t_1 , so they still have the money they earned since the beginning of the iteration). Analogously as in the preceding paragraph, projects $d_{i+1}, \dots, d_n$ also are supported by voters who can afford them at time $t_1 + t_2$ but d_i wins due to the tie-breaking order (and, assuming that p were not selected yet, the other D -projects were purchased in previous rounds).

At time $t_1 + t_2$ since the beginning of the iteration, each of the not-yet-selected universe projects is supported by voters who in total have at most:

$$z \cdot t_2 + s \cdot (n - 3) \cdot t_2 + 3s \cdot t_1 < 1.$$

money, so neither of them can be purchased at this time.

Finally, let us consider the amount of money that voters supporting p have at time $t_1 + t_2$. To this end, let g_i be the number of set voters whose corresponding sets contain u_i (and, hence, who did not spend their money since the beginning of the iteration) and who also got additional approvals for p . Consequently, at time $t_1 + t_2$ since the beginning of the iteration, voters approving p have the following amount of money (assuming that p was not purchased prior to this time):

$$f(g_i) = z \cdot t_2 + \ell \cdot t_2 + \alpha \cdot (t_1 + t_2) + g_i \cdot t_1.$$

Indeed, each Z -voter earned t_2 since time t_1 , each set voter who got an approval for p also earned t_2 since time t_1 , each A -voter earned $t_1 + t_2$ since the beginning of the iteration, and each of the g_i set voters who got an approval for p and who corresponds to a set including u_i did not spend its t_1 amount of money at time t_1 . We observe that:

$$f(1) < 1 \leq f(2).$$

Hence, if $g_i = 1$ then PHRAGMÉN selects d_i at time $t_1 + t_2$ since the beginning of the iteration, and we start iteration $i + 1$ (with each Z -voter having t_2 money and each Y and each set voter having zero money). Yet, if $g_i \geq 2$, then PHRAGMÉN selects p (possibly even earlier than time $t_1 + t_2$, especially if $g_i > 2$).

All in all, it is possible to add ℓ approvals for p to the set voters without getting p to be funded if and only if our input RX3C instance has an exact cover.

Adding p -Approvals Beyond Set Voters. Next, we explain why it suffices to focus on adding p -approvals to the set voters. We make the following observations:

1. If we add a p -approval to even a single W -voter (and the remaining $\ell - 1$ approvals to whatever other voters), then at latest at time $t_1 + t_2$ since the beginning of the second iteration, this voter would have accumulated sufficient amount of money that p would be selected instead of d_2 .
2. All Z -voters already approve p , so it is impossible to add further p -approvals to them.
3. If there is a solution where we add p -approvals to some Y -voters (who are not A -voters, as those already approve p), then p also is not funded if we add the approvals to set voters instead. Indeed, the Y -voters have at least as much money as the set voters at each point of time during each iteration.

This justifies why it suffices to consider adding approvals for p to the set voters only. However, regarding the Y -voters we can say something stronger: If we add an approval to some Y -voter y_j , then in each iteration i , when considering d_i vs p , voter y_j would contribute $t_1 + t_2$ to buying p (since y_j does not pay for u_i), whereas any S -voter s_k contributes $t_1 + t_2$ only if u_i belongs to the corresponding set (otherwise s_k contributes to u_i and is left with $t_2 < t_1 + t_2$ budget for p). Thus, in iteration i at time $t_1 + t_2$ p -supporters would have budget at least:

$$z \cdot t_2 + \ell \cdot t_2 + \alpha \cdot (t_1 + t_2) + g_i \cdot t_1 + y_p \cdot t_1,$$

where $y_p \geq 1$ is the number of Y -voters that got approvals for p (g_i was defined when analyzing the second part of the iteration). It is not hard to see that $y_p \geq 2$ results in selecting p before d_1 and $y_p = 1$ results in selecting p before d_h , where h is the index of element u_h whose one of the voters corresponding to sets containing u_h received an approval towards p . Therefore, adding a p -approval to any of the Y -voters (who is not a A -voter) results in selecting p .

Consequently, if there is a solution for our problem, then it consists of adding p -approvals to set voters only. This means that the number of solutions for our problem is equal to the number of exact covers in the input RX3C instance.

Calculations. To complete the proof, we establish that the (in)equalities that we asserted actually hold.

1. $1 = (y + s \cdot n + w) \cdot t_2/2$. This equation asserts that after time $t_2/2$, candidate b_1 can be bought (and later analogously candidate b_2). This is true because $(y + s \cdot n + w) \cdot t_2/2 = (\frac{4n^{10}-3n^9}{18} + n^9/6 \cdot n + \frac{11n^{10}+3n^9}{18}) \cdot 1/n^{10} = \frac{4n^{10}-3n^9+3n^{10}+11n^{10}+3n^9}{18n^{10}} = \frac{18n^{10}}{18n^{10}} = 1$.
2. $1 = z \cdot (t_1 + t_2) + s \cdot (n - 3) \cdot t_1$. This equation asserts that U -candidate u_i can be selected after time t_1 provided that Z -voters had initially budget t_2 . One can verify that $\frac{1-s \cdot (n-3) \cdot t_1}{t_1+t_2} = \frac{n^{10}-n^9/6 \cdot (n-3) \cdot 1}{1+2} = \frac{5n^{10}+3n^9}{18} = z$.
3. $1 = y \cdot (t_1 + t_2) + s \cdot n \cdot t_2 + 3 \cdot s \cdot t_1$. This equation asserts that D -candidate d_i can be selected after time $t_1 + t + 2$ from the beginning of an iteration provided that formerly U -candidate u_i was selected at time t_1 . One can verify that $\frac{1-s \cdot n \cdot t_2 + 3 \cdot s \cdot t_1}{t_1+t_2} = \frac{n^{10}-n^9/6 \cdot n \cdot 2 - n^9/6 \cdot 3 \cdot 1}{1+2} = \frac{4n^{10}-3n^9}{18} = y$. It also implies that $1 - z \cdot (t_1 + t_2) = s \cdot (n - 3) \cdot t_1$.
4. $z \cdot t_2 + \ell \cdot t_2 + \alpha \cdot (t_1 + t_2) + t_1 < 1 \leq (z \cdot t_2 + \ell \cdot t_2 + \alpha \cdot (t_1 + t_2) + t_1) + t_1$. This inequality asserts for each U -candidate u_i that if we add approval to one voter corresponding to a set containing u_i , then p still loses with d_i at time t_2 , but if we add approvals to at least two voters corresponding to sets containing u_i , then p will be selected (due to tie-breaking at $t_1 + t_2$ for two or sooner for at least three). Please note that it is equivalent to saying that $\alpha \in [\frac{1-(z \cdot t_2 + \ell \cdot t_2 + 2t_1)}{t_1+t_2}, \frac{1-(z \cdot t_2 + \ell \cdot t_2 + t_1)}{t_1+t_2})$. Since $\alpha = \frac{4n^{10}-3n^9-6n-18}{27} = \frac{9n^{10}-5n^{10}-3n^9-6n-18}{27} = \frac{n^{10}-\frac{5n^{10}+3n^9}{18} \cdot 2 - n/3 - 2}{1+2} = \frac{1-z \cdot t_2 - n/3 \cdot t_2 - 2t_1}{t_1+t_2}$, α lies in this range.
5. $y \cdot t_1 + s \cdot n \cdot t_1 < 1$. This inequality guarantees at in iteration i at time t_1 , each yet-unbought D -candidate will have too low budget of its supporters to be bought so we will prefer to purchase u_i . This also holds as $y \cdot y_1 + s \cdot n \cdot t_1 = \frac{\frac{4n^{10}-3n^9}{18} + n^9/6 \cdot n}{n^{10}} = \frac{4n^{10}-3n^9+3n^{10}}{18n^{10}} = \frac{7n^{10}-3n^9}{18n^{10}} < 1$.
6. $z \cdot (t_1 + t_2) + \ell \cdot t_1 + \alpha \cdot t_1 < 1$. This inequality assures that in iteration i at time t_1 , p will have too low budget of its supporters to be bought so we will prefer to purchase u_i . Let us observe that $z \cdot (t_1 + t_2) + n/3 \cdot t_1 + \alpha \cdot t_1 < 1 \iff n/3 \cdot t_1 + \alpha \cdot t_1 < 1 - z \cdot (t_1 + t_2) \iff n/3 \cdot t_1 + \alpha \cdot t_1 < s \cdot (n - 3) \cdot t_1 \iff n/3 + \alpha < s \cdot (n - 3) \iff s \cdot (n - 3) - n/3 - \alpha > 0$. Since $s \cdot (n - 3) - n/3 - \alpha = n^9/6 \cdot (n - 3) - n/3 - \frac{4n^{10}-3n^9-6n-18}{27} = \frac{9n^{10}-27n^9-18n-8n^{10}+6n^9+12n+36}{54} = \frac{n^{10}-21n^9-6n+36}{54} = \frac{(n-22) \cdot n^9 + (n^8-6) \cdot n + 36}{54} > 0$ for $n \leq 60$, this inequality also holds.

7. $z \cdot t_2 + s \cdot (n-3) \cdot t_2 + 3 \cdot s \cdot t_1 < 1$. This inequality asserts that in iteration i at time $t_1 + t_2$, each yet-unbought U -candidate will have too low budget of its supporters to be bought so we will purchase d_i or p . If we place the actual values in the left side, we obtain that
- $$z \cdot t_2 + s \cdot (n-3) \cdot t_2 + 3 \cdot s \cdot t_1 = \frac{\frac{5n^{10}+3n^9}{18} \cdot 2 + n^9/6 \cdot (n-3) \cdot 2 + 3 \cdot n^9/6 \cdot 1}{n^{10}} = \frac{10n^{10}+6n^9+6n^{10}-18n^9+9n^9}{18n^{10}} = \frac{16n^{10}-3n^9}{18n^{10}} < 1.$$

This completes the proof. $\square$

Theorem 3.6. *For PH and EQ, there is an algorithm that computes $\text{pessimist-add}_E(p)$ and runs in FPT time with respect to parameter $|f(E)|$, i.e., the number of rounds.*

Proof. We start with PHRAGMÉN and later show how the algorithm can be tweaked to work also for EQUAL-SHARES. Let $E = (P, V, B)$ be a PB instance with losing project p . We will show how to compute the largest number of approvals whose addition *does not* lead to funding p . The value of $\text{pessimist-add}_E(p)$ is one larger. Let $k = |\text{PHRAGMÉN}(E)|$ be the number of funded projects or, equivalently, the number of rounds performed by the rule. We assume that p is approved by at least one voter in V .

For each round i , let m_i be the difference between $\text{cost}(p)$ and the amount of funds that voters approving p have in round i . Our goal is to select the largest group of voters who do not approve p and whose total funds in each round i are at most m_i .

For each voter $v_i \in V \setminus A(p)$, we define its *balance vector* $(b_1^i, \dots, b_k^i)$, which contains the balance of the voter's bank account right before each round. We partition the voters not approving p into *voter-types* $T = \{T_1, \dots, T_t\}$, where each type consists of voters with identical balance vectors.

To solve the problem, we form an integer linear program (ILP). For every voter-type T_i we have a nonnegative integer variable x_{T_i} that represents the number of voters of this type that will additionally approve p . For each round i we form the following *round-constraint* (using strict inequality, if needed due to tie-breaking):

$$\sum_{j \in \{1, \dots, t\}} x_{T_j} \cdot b_i^{T_j} \leq m_i.$$

The objective function is to maximize the sum of all the x_{T_i} variables. Our algorithm outputs the value of the objective function plus one.

Finally, we note that there are at most $O(2^k)$ voter types (indeed, each voter type corresponds to a k -dimensional 0/1 vector, which has 1 in position i if a voter approves—and, hence, pays for—the candidate selected in round i or not). Thus, the number of variables in our ILP is $O(2^k)$ and we can solve it using the classic algorithm of [Lenstra, Jr. \[1983\]](#) in FPT time with respect to k .

For EQUAL-SHARES, the algorithm is almost the same. We just extend the preprocessing by computing values $q_1, \dots, q_k$ that stand for the value of $q(c_i)$ used to fund project c_i in round $i \in [k]$ and we additionally extend the definition of balance vector by the value b_{k+1}^i . In the ILP, we then replace each round-constraint for each round $i \in [k]$ with a constraint:

$$\sum_{j \in [t]} x_{T_j} \cdot \min(b_i^{T_j}, q_i \cdot \text{cost}(p)) \leq m_i,$$

which ensures that p is not funded instead of c_i in round i (again, assuming p is last in the tie-breaking order). The only missing component is to secure that p is not selected as the $(k+1)$ -th funded project (as p had at least one approval to begin with, such a situation was impossible under PHRAGMÉN). To prevent this, we add a final constraint

$$\sum_{j \in [t]} x_{T_j} \cdot b_{k+1}^{T_j} \leq m_{k+1}.$$

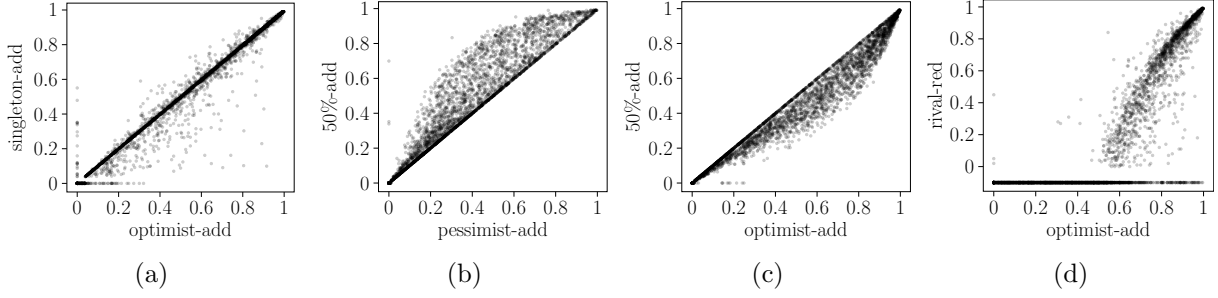


Figure 6: Further correlation plots between our measures where each point is one project for EQ/PHRAGMÉN (continues Figure 1). Measures are normalized so that 1 denotes no change and 0 denotes a maximal-size change. In Figure 6 (d), a negative value of rival-red means that for all considered values of ℓ removing rivalry approvals from ℓ supporters selected uniformly at random was not sufficient for a 50%-funding probability.

The rest of the arguments and the running time remain the same as for PHRAGMÉN. $\square$

B.4. Adding-Singletons Measure

Proposition 3.8. *For AV and PH, $\text{singleton-add}_E(p)$ can be computed alongside the rule, at an $O(1)$ cost per round.*

Proof. In this proof, we assume that the $\text{singleton-add}_E(p)$ measure is well-defined, which can be easily checked in constant time before we run our algorithm; see Appendix B.1.

The rough idea for all the rules is very the same. We simulate the execution of a respective PB rule and, in each round, we compute the minimum number of voters approving p only that need to be added to the instance such that p is funded in this round. The value of $\text{singleton-add}_E(p)$ is then the minimum value over all rounds.

We start with the AV rule. The algorithm is very the same as in Proposition 3.4. We run the rule on the original instance as usual. The interesting part of the process takes place once the funding of the currently processed project c first decreases the value of the remaining budget below $\text{cost}(p)$. Before funding c , we compute m^* as $|A(c)| - |A(p)| + [c \succ p]$, where $[c \succ p]$ evaluates to 1 if c precedes p in tie-breaking order and to 0 otherwise. The algorithm then outputs the value of m^* . For correctness, suppose that $\text{singleton-add}_E^{\text{AV}}(p) = m' < m^*$. If we add m' additional voters approving p , then p is assumed by the rule in some round after c is funded. However, c was selected so that, after funding c , the remaining budget is below $\text{cost}(p)$. Consequently, adding m' additional voters is not sufficient to fund p and therefore $\text{singleton-add}_E^{\text{AV}}(p) \geq m^*$. Since, by the definition of the rule, the distinguished project p is funded after adding m^* approvals, the algorithm is correct and clearly performs only $O(1)$ additional operations per round.

Next, let the rule be PH. We again simulate the rule and, this time, in every round i , we compute the minimum number of voters who approve solely p that we need to add to the instance to make p funded in round i . Let c be a project originally funded in round i , b_i be the budget of a virtual voter approving only p in this round, and e_i^p be the sum of endowments of all voters who approve p just before round i in the original instance. To make p funded, we need to ensure that $m_i \cdot b_i + e_i^p \geq \text{cost}(p)$, where $m_i \in \mathbb{N}$ is the number of added voters. If $m_i \cdot b_i + e_i^p = \text{cost}(p)$, the tie-breaking prefers c over p , and there exists at least one voter who approves both p and c , we need to make the inequality strict. We stop the execution once the remaining budget is lower than $\text{cost}(p)$ and the algorithm outputs the minimum m_i over all rounds.

To conclude, observe that the computation of m_i is $O(1)$ operation along the standard computation of the rule. Moreover, we can create one “global” variable m^* to store the minimum

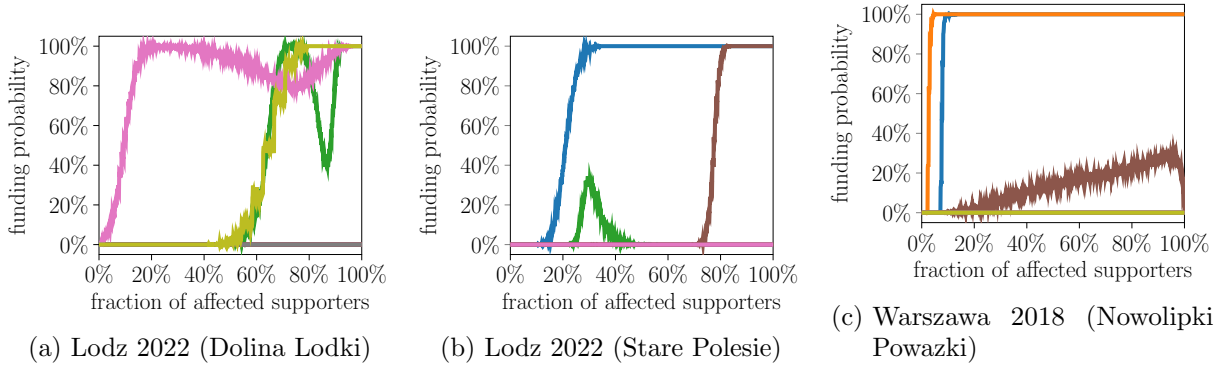


Figure 7: Line plots showing how the funding probability of a project develops if we remove rivalry approvals from its supporters selected uniformly at random for EQ/PHRAGMÉN (continues Figure 4).

m_i . The variable is originally initiated to ∞ , and in each round we conditionally update it if we face a smaller value than the one currently stored. This also requires a constant number of additional operations per round, and the theorem follows. $\square$

B.5. Rivalry-Reduction Measure

Proposition 3.10. *For AV, PH, and EQ, the problem of deciding, given a PB instance E , a losing project p , and an integer ℓ , if it is possible to ensure p 's victory by changing at most ℓ votes that originally approve p to only approve p is NP-complete, even when $B = 1$ and every project costs 1.*

Proof. Membership to NP is clear for all three rules: given a set of voters $S \subseteq V$ to be adjusted as a certificate, we can verify that this is indeed a solution by checking whether all of them approve p and by setting $A(v) = p$ for every $v \in S$ and running the rule on the modified instance. Since all the rules run in polynomial-time, the membership holds.

To prove NP-hardness under the AV rule, we reduce from the SET-COVER problem, which is known to be NP-complete Karp [1972]. Let $\mathcal{I} = (U, S, k)$ be an instance of SET-COVER, where U is a set of elements (universe), S is a collection of sets of elements from U , and k is an integer. Our goal is to decide whether there exist k -sized set $S' \subseteq S$ such that each element of U appears in at least one of the selected sets.

We construct an equivalent instance $\mathcal{J}$ of our problem as follows. We create one *universe-project* for every element $u \in U$ and add one distinguished project p . All the projects have cost 1, and we have $|P| = |U| + 1$. For each set $S_j \in S$, we create one *set-voter* approving p and candidates corresponding to elements of S_j . Furthermore, for each project $u_i \in U$, we create $|S| - |S(u_i)| + 1$, where $S(u_i)$ is the set of sets containing u_i as an element, *dummy-voters* who approve only u_i . That is, each universe-project is approved by $|S| + 1$ voters and p is approved by exactly $|S|$ voters. To complete the construction, we set the budget $B = 1$, the number of votes to change $\ell = k$, and the tie-breaking order is $p \succ u_1 \succ u_2 \succ \dots \succ u_{|U|}$. Observe that exactly one project will be funded, as all project are unit-cost and the budget is $B = 1$.

For the correctness, let $\mathcal{I}$ be a yes-instance and $S' \subseteq S$ be a set cover of size k . We remove rivalry approvals of all set-voters corresponding to the sets in S' . Since S' is a set cover, every universe project loses at least one approval. Therefore, each universe project is approved by at most $|S|$ voters and so is the distinguished candidate p . Therefore, due to our tie-breaking order, in the modified instance p will be selected as the only funded candidate and, thus, voters corresponding to elements of S' are a solution for $\mathcal{J}$.

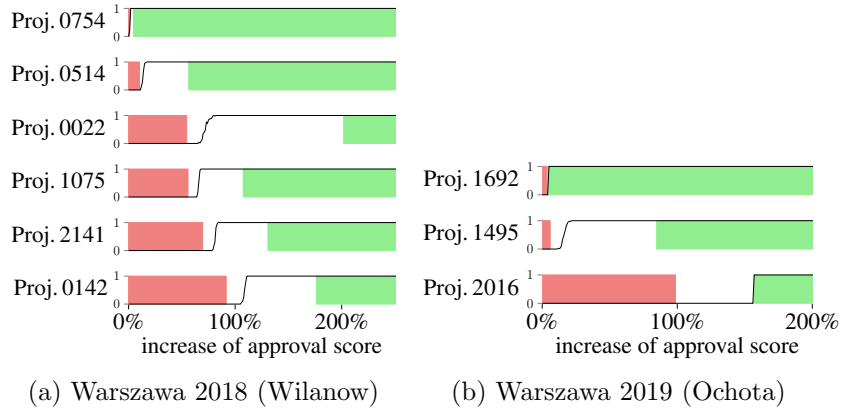


Figure 8: Line plots showing how the funding probability of a project develops from 0 to 1 when increasing its approval score by adding approvals uniformly at random to existing voters for EQ/PHRAGMÉN (continues Figure 2). The red area goes until the optimist-add value and the green area extends from the pessimist-add value.

In the opposite direction, let $\mathcal{J}$ be a yes-instance and $V' \subseteq V \cap A(p)$ be an ℓ -sized subset of voters such that setting $A(v) = \{p\}$ for every $v \in V'$ leads to the funding of p . Clearly, the set V' consists exclusively of set-voters, as no dummy-voter approves p . Since by removing rivalry approvals of voters in V' the distinguished candidate become funded, every universe-project loses at least one approval. In other words, the removed approvals contain each universe-project at least once; setting S' to be the set of all sets in S corresponding to set-voters in V' , we obtain $|S'| = |V'| = \ell = k$ and that S' is a set cover in $\mathcal{I}$ by the previous argumentation.

The reduction can be clearly done in polynomial time and the NP-completeness for AV follows. Surprisingly, we can also use the same construction to prove NP-completeness for PH and EQ. Observe that there is only one round and all the rules select the most approved candidate as the single winner. Hence, the problem is NP-complete for all rules of our interest. $\square$

C. Additional Material for Section 4

C.1. Additional Plots for Eq/Phragmén

Figure 6 shows additional correlation plots for pairs of our measures. Figure 7 includes further examples of instances where projects show a non-monotonic behavior when removing rivalry approvals from project's supporters uniformly at random. Lastly, in Figure 8, we give some further examples of instances where some projects have a substantial gap between the optimist-add and pessimist-add values yet their funding probabilities perform the typical jumps when adding approvals to existing voters uniformly at random.

C.2. Analysis for Phragmén

In this section, we describe the results of our experimental analysis for PHRAGMÉN. The general picture is very similar as for EQ/PHRAGMÉN and our presented figures are created analogous to the ones for PHRAGMÉN. In Table 3, we show the Pearson Correlation Coefficient (PCC) between our measures and in Figure 9 correlation plots for some of them. The PCC values for PHRAGMÉN are very similar to the ones for EQ/PHRAGMÉN, typically differing by at most 0.02. The only larger differences appear for rival-red, which has for PHRAGMÉN a higher correlation to the other measures than for EQ/PHRAGMÉN. (Note that for PHRAGMÉN 1484 out of the 3513

	optimist	pessimistic	50%	singleton	rival	cost
optimist	–	0.89	0.98	0.98	0.85	0.74
pessimistic	0.89	–	0.96	0.88	0.76	0.79
50%	0.98	0.96	–	0.96	0.85	0.79
singleton	0.98	0.88	0.96	–	0.93	0.78
rival	0.85	0.76	0.85	0.93	–	0.70
cost	0.74	0.79	0.79	0.78	0.70	–

Table 3: Pearson Correlation Coefficient between measures for PHRAGMÉN.

projects have a funding probability above 50% for some considered value of ℓ when removing rivalry approvals from ℓ of its supporters uniformly at random). Also, the correlation plots for PHRAGMÉN look very similar to the ones for EQUAL-SHARES. The most significant difference here is probably in [Figure 9 \(a\)](#) which compares singleton-add and optimist-add, which is due to the fact that as discussed in the main body for PHRAGMÉN singleton-add constitutes a lower bound for optimist-add, which is not the case for EQ/PHRAGMÉN.

Regarding 50%-add, we again observe that the measure is slightly closer to optimist-add than to pessimist-add. The average (resp., maximum) difference between 50%-add and optimist-add is 0.07 (resp., 0.3), whereas it is 0.1 (resp., 0.4) for the 50%-add measure and pessimist-add. Again we find that even in case there is a gap between the optimist-add and pessimist-add value of a project, project’s funding probability quickly transitions from an almost 0% to a close-to 100% when adding approvals to existing voters uniformly at random. [Figure 10](#) shows the behavior of several instances in support of this claim.

Regarding rival-red, for PHRAGMÉN there are more projects whose funding probability behaves non-monotonically when removing rivalry approvals than for EQ/PHRAGMÉN. Again we present in [Figure 11](#) some cherry-picked instances where the funding probabilities of some projects show a particularly interesting behavior.

C.3. Analysis for greedyAV

We repeated our experiments for GREEDYAV where 3581 projects ended up being not funded. Notably, here all measures involving adding approvals to existing voters or adding singleton voters coincide, so we only analyze cost-red, singleton-add, and rival-red.

[Figure 12](#) shows correlation plots and PCC values for these three pairs of measures. Regarding the relationship between singleton-add and cost-red ([Figure 12 \(a\)](#)), the connection between the two is much weaker than for the proportional rules. In particular, there are many projects with a small value of cost-red and varying values of singleton-add. This can be explained quite easily by recalling the inner workings of GREEDYAV: When modifying only project’s costs, the ordering in which projects are considered remains unchanged. Thus, a project can only cost the amount of money left when it is their turn in the original ordering of projects. Now, in case most or all of the budget has already been spent before, there is no money for the project left. In contrast, modifying the approval score of a project allows us to influence its position in the order in which projects are considered and thereby how much money is left for the project once it is its turn. Accordingly, the difference to the proportional rules here is due to the fact that for GREEDYAV, cost-red is in some sense of more limited power.

For rival-red, for 862 projects for some considered value of ℓ removing rivalry approvals from ℓ supporters selected uniformly at random was sufficient for a 50% funding probability. Notably, in contrast to the proportional rules, removing rivalry approvals has a different, arguably weaker effect for GREEDYAV, as it only reduces the approval score of competing projects and not how much of their preallocated money voters have left to spend on the designated project (because

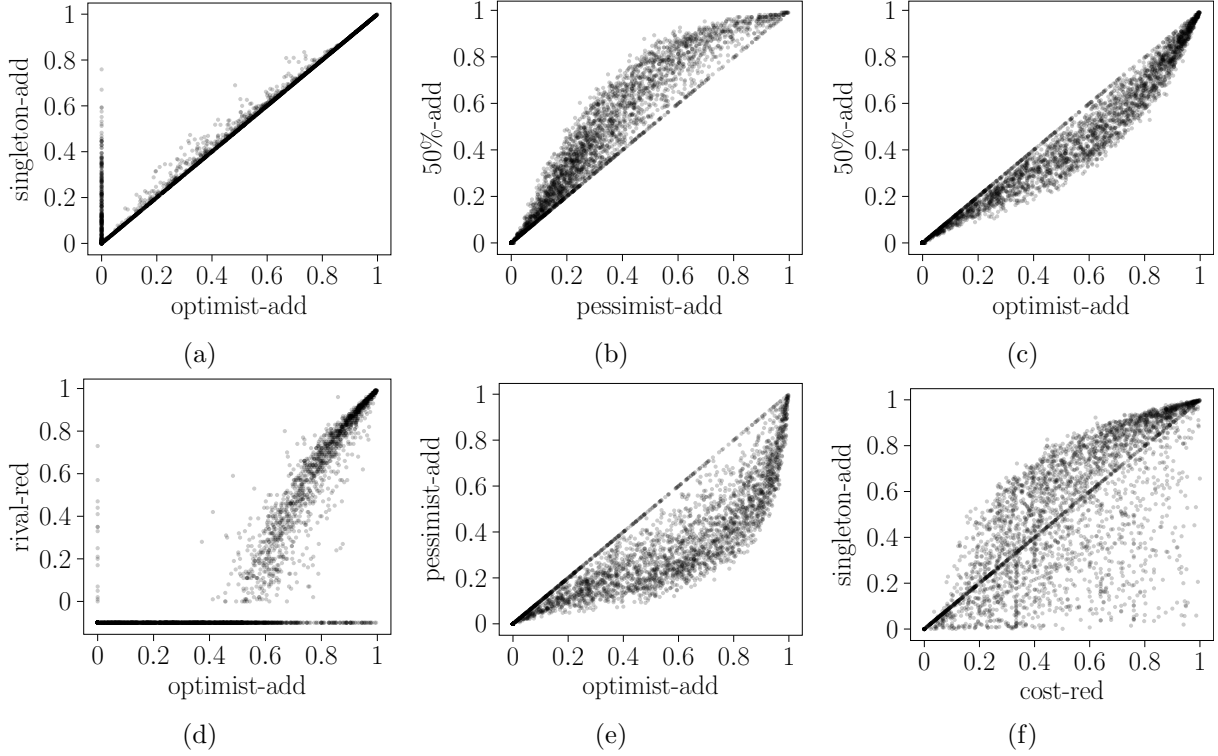


Figure 9: Correlation plots where each point is one project for PHRAGMÉN. A negative value of rival-red means that for all considered values of ℓ removing rivalry approvals from ℓ supporters selected uniformly at random was not sufficient for a 50%-funding probability.

there is no preallocated money for GREEDYAV). Thus, it is to be expected that rival-red is less powerful for GREEDYAV than for the proportional rules. In fact, the fact that rival-red has an impact on 862 projects can be even regarded as a surprisingly high number, highlighting that also for GREEDYAV adding additional approvals to one’s ballot can hurt the funding possibilities of other approved projects.

Remarkably, for GREEDYAV, there are even more projects than for the proportional rules whose funding probability behaves non-monotonically when removing rivalry approvals and the behavior of these projects comes in more different flavors. Figure 13 shows some selected instances. One instance that sticks out in particular is Figure 13 (e), where the gray project has an almost 100% funding probability when few of its supporters remove their rivalry approvals; however, in case some more do it as well its funding probability drops again to 0% and even in case gray’s supporters don’t approve any other projects, the project does not get funded again. Nevertheless, for a majority of projects, their funding probability does behave monotonically and quickly jumps from around 0% to around 100% (the instances shown in Figure 13 are meant to demonstrate the non-monotonic behavior and not to provide representative coverage of the project’s behavior). Examining the relationship of rival-red to the other two measures in more detail, we see almost no positive correlation with cost-red in Figure 12 (b). In contrast, the connection to singleton-add (see Figure 12 (c)) is stronger. The correlation (plot) of singleton-add and rival-red is quite similar to the respective plots for PHRAGMÉN (Figure 9 (f)) and EQ/PHRAGMÉN (Figure 6 (d)). The above observations indicate that also the power of rival-red for PHRAGMÉN and EQ/PHRAGMÉN stems only partly from the proportionality of the two rules, yet also from the fact that there is simply more money left in case competing projects are not funded.

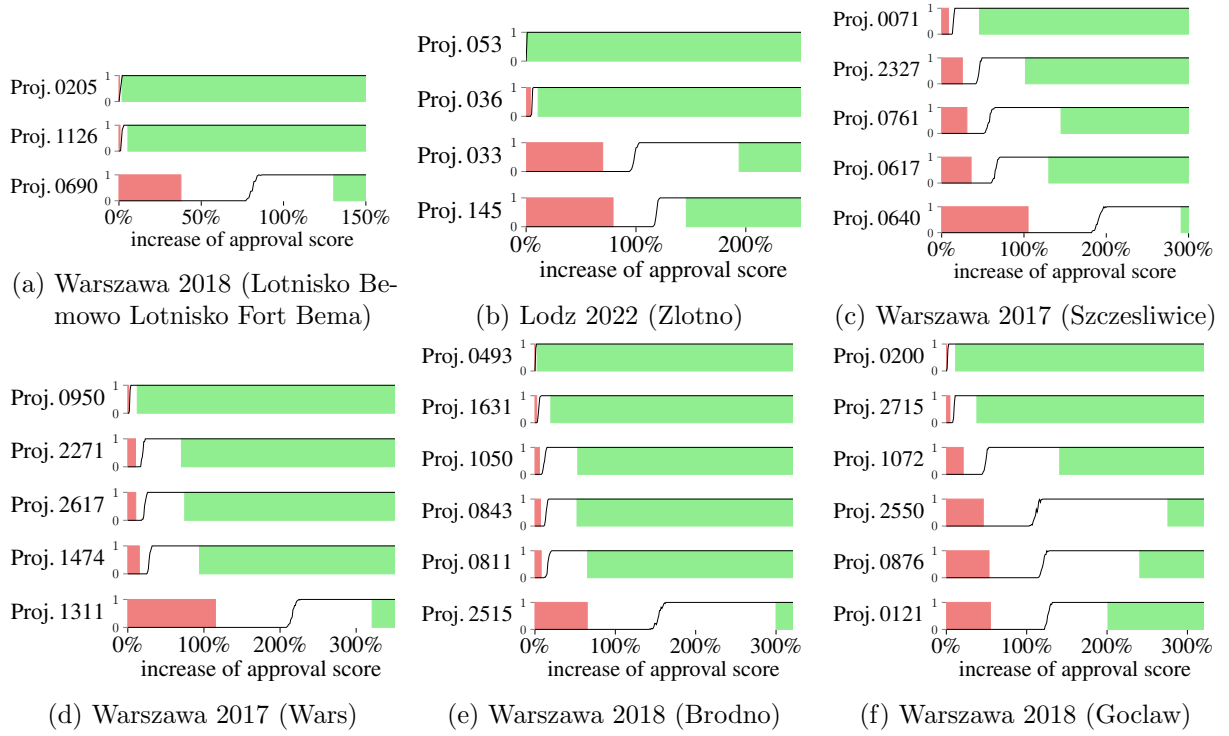


Figure 10: For PHRAGMÉN, line plots showing how the funding probability of projects develops from 0 to 1 when increasing their approval scores by adding approvals uniformly at random to existing voters. The red area goes until the optimist-add value and the green area extends from the pessimist-add value.

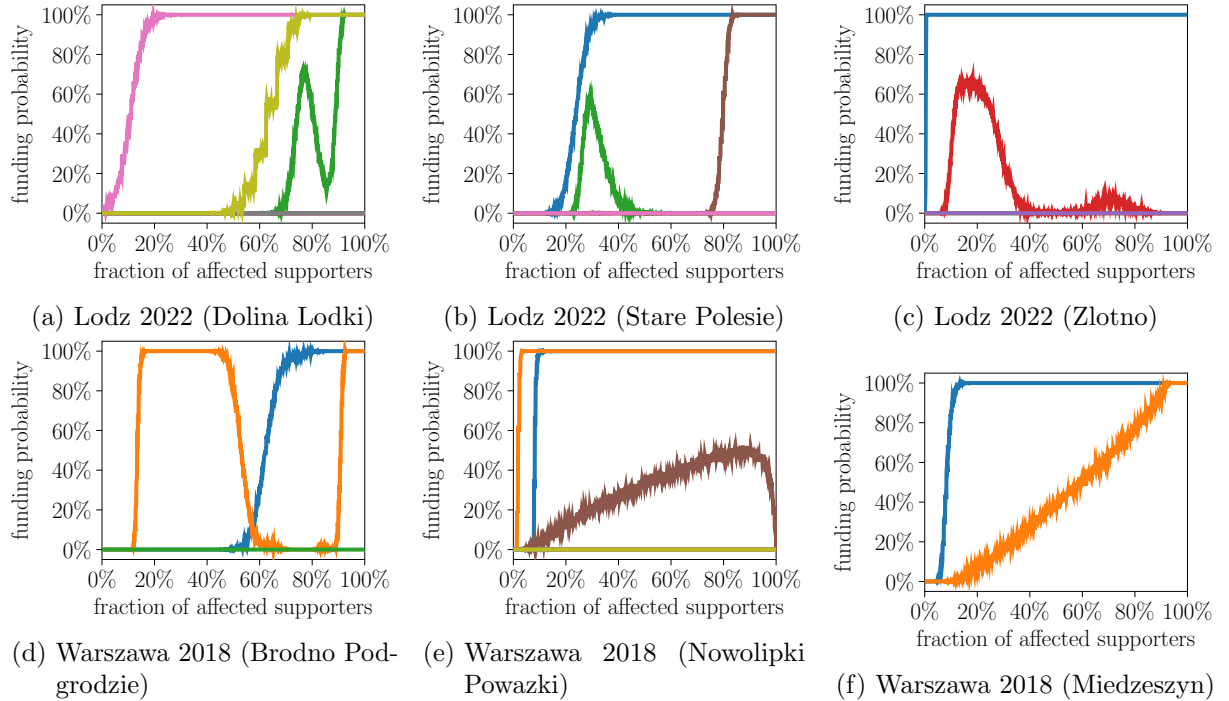


Figure 11: For PHRAGMÉN, line plots showing how the funding probability of projects develops if we remove rivalry approvals from subgroups of their supporters selected uniformly at random.

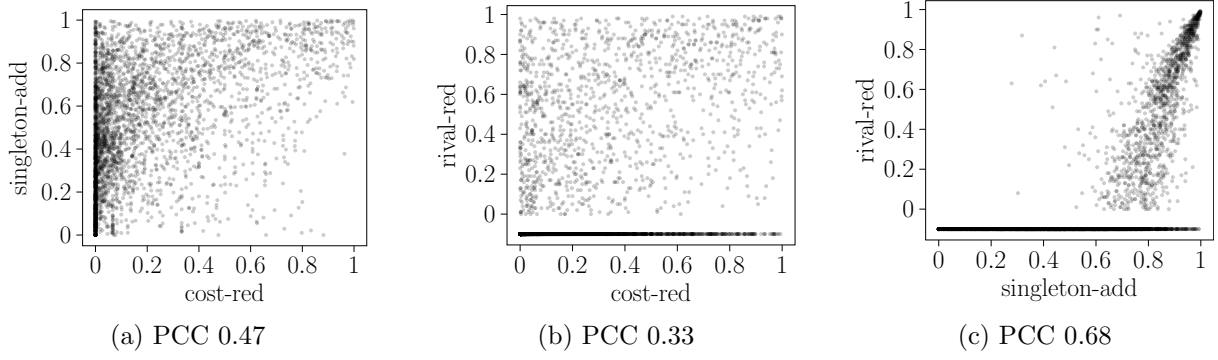


Figure 12: For GREEDYAV, correlation plots where each point is one project. A negative value of rival-red means that for all considered values of ℓ removing rivalry approvals from ℓ supporters selected uniformly at random was not sufficient for a 50%-funding probability.

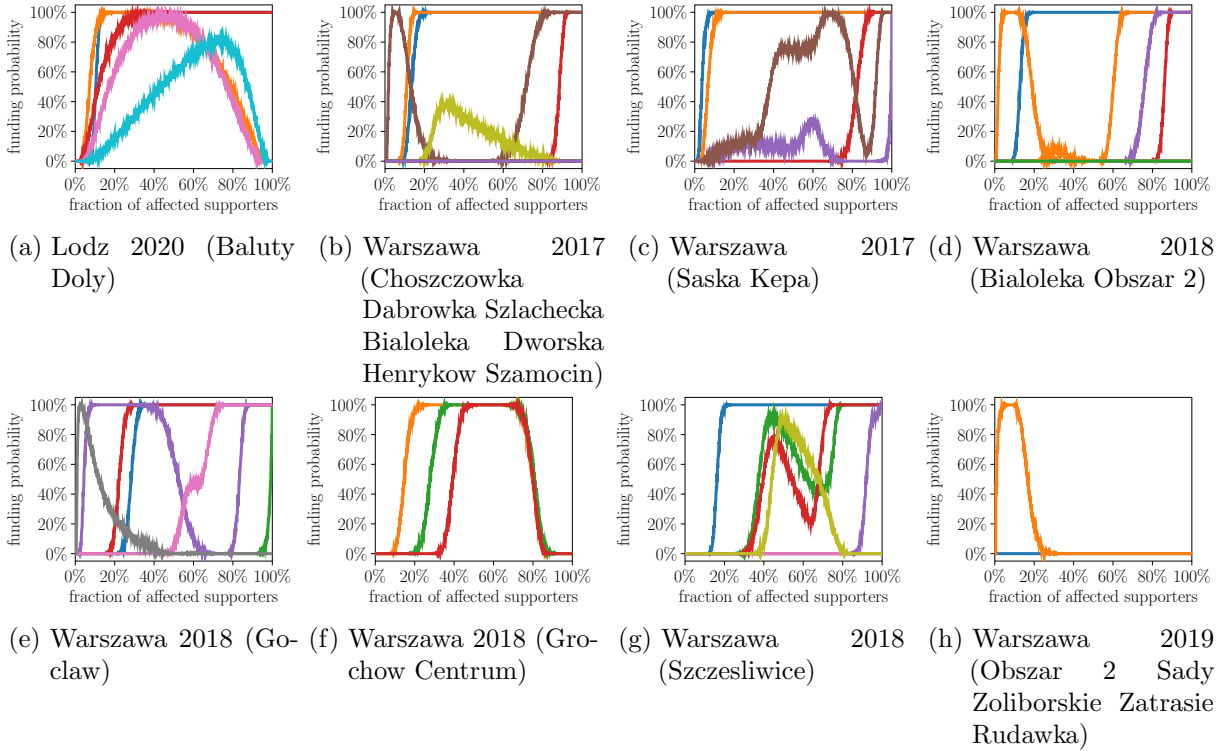


Figure 13: For GREEDYAV, line plots showing how the funding probability of projects develops if we remove rivalry approvals from subgroups of their supporters selected uniformly at random.